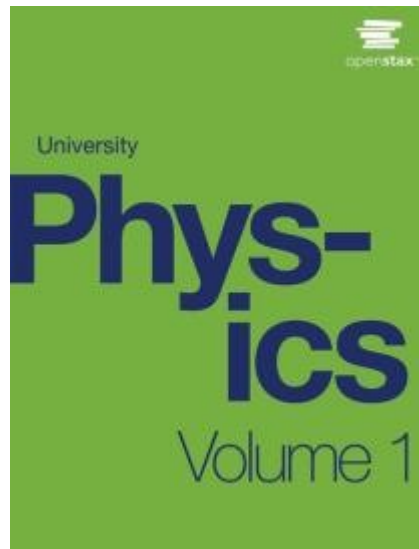


# UNIVERSITY PHYSICS

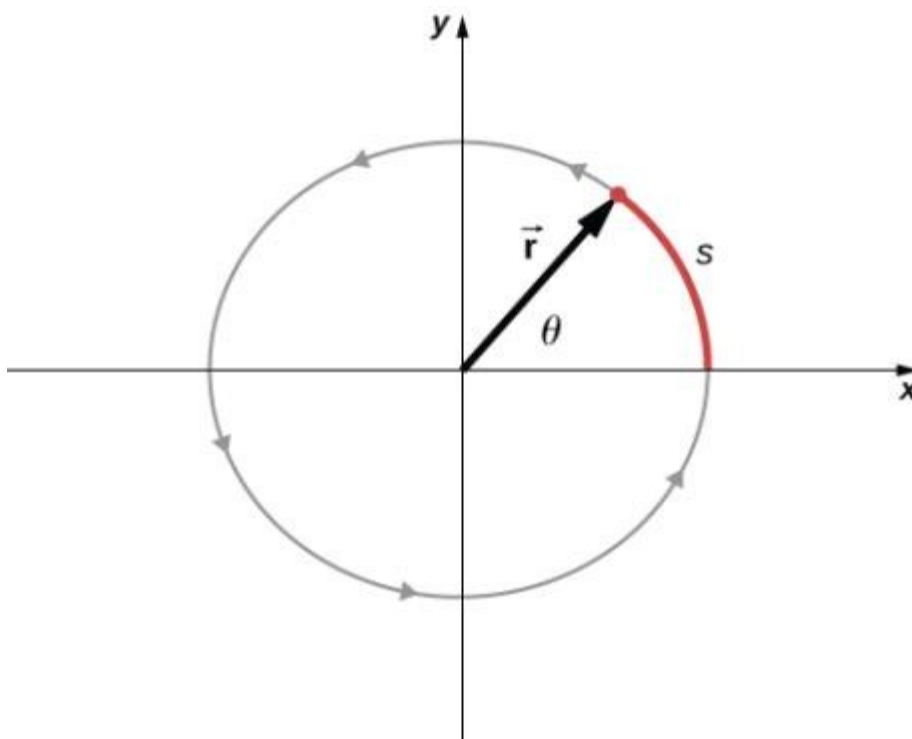
## Chapter 10 FIXED-AXIS ROTATION

PowerPoint Image Slideshow



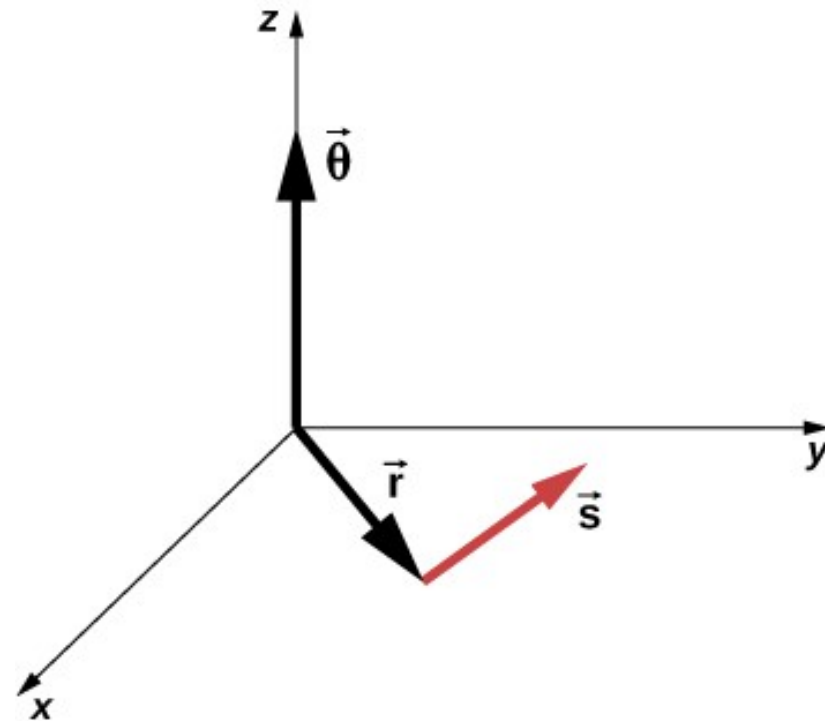
# **Coordinate systems & terminology**

**FIGURE 10.2**



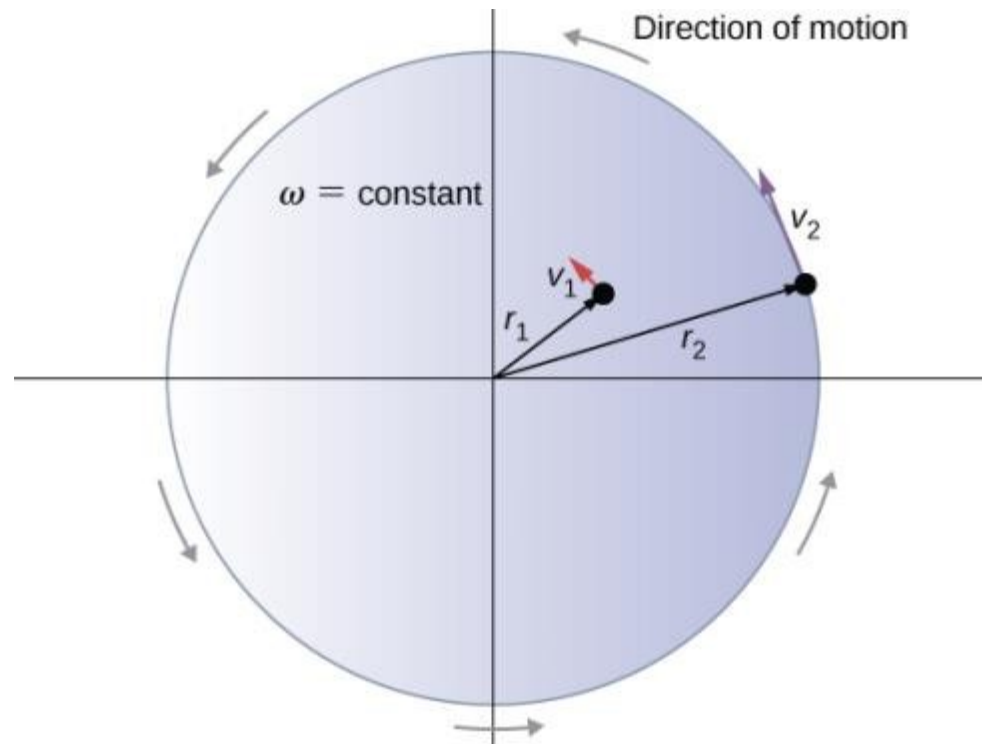
A particle follows a circular path. As it moves counterclockwise, it sweeps out a positive angle  $\theta$  with respect to the x-axis and traces out an arc length  $s$ .

# FIGURE 10.3



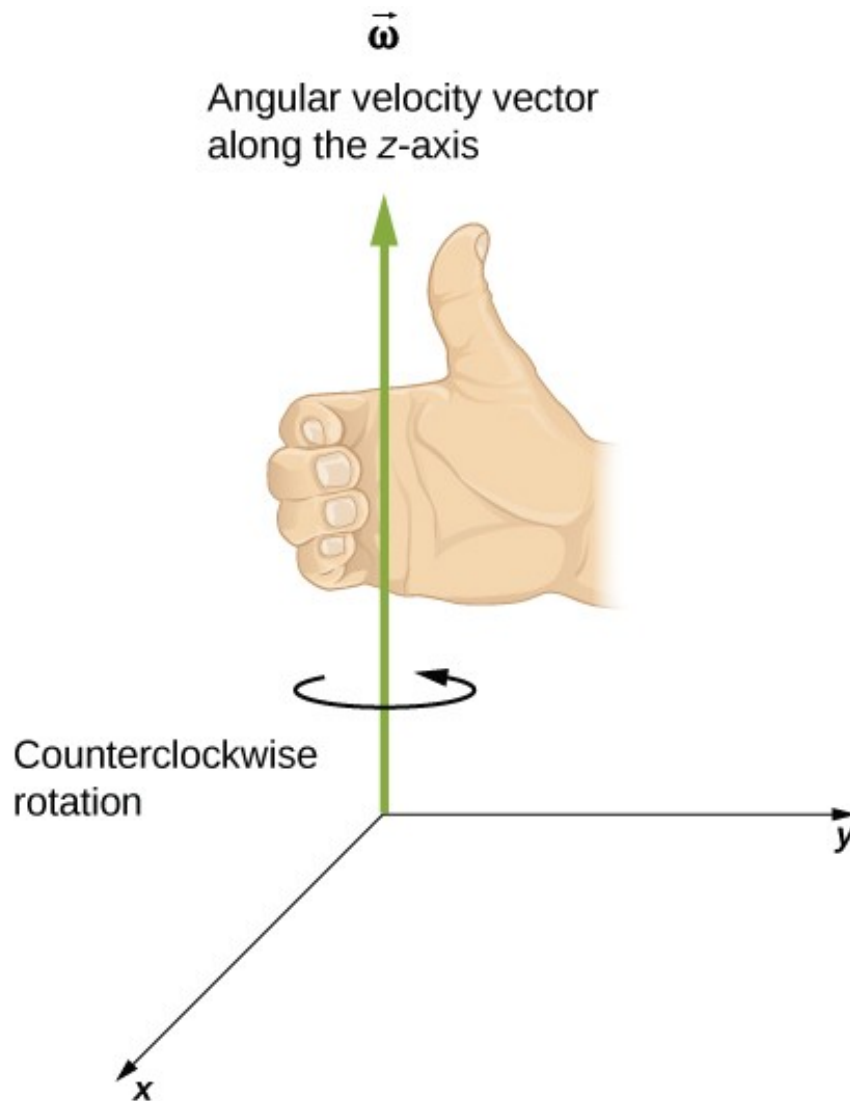
The angle vector points along the  $z$ -axis and the position vector and arc length vector both lie in the  $xy$ -plane. We see that . All three vectors are perpendicular to each other.

**FIGURE 10.4**



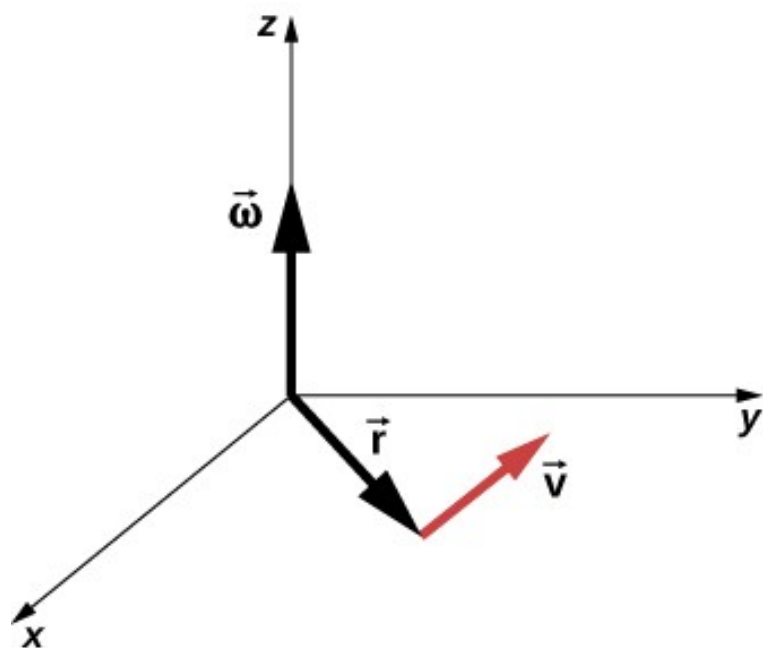
Two particles on a rotating disk have different tangential speeds, depending on their distance to the axis of rotation.

## FIGURE 10.5

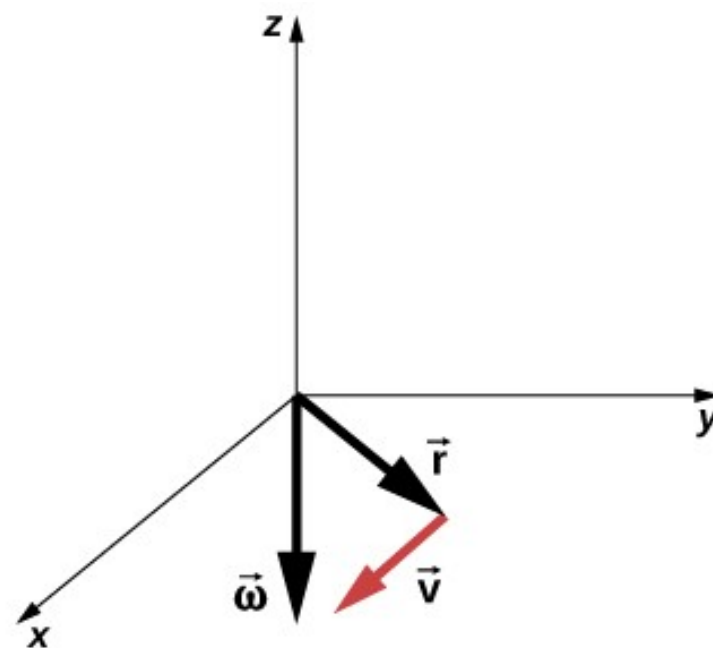


For counterclockwise rotation in the coordinate system shown, the angular velocity points in the positive z-direction by the right-hand-rule.

## FIGURE 10.6



(a)

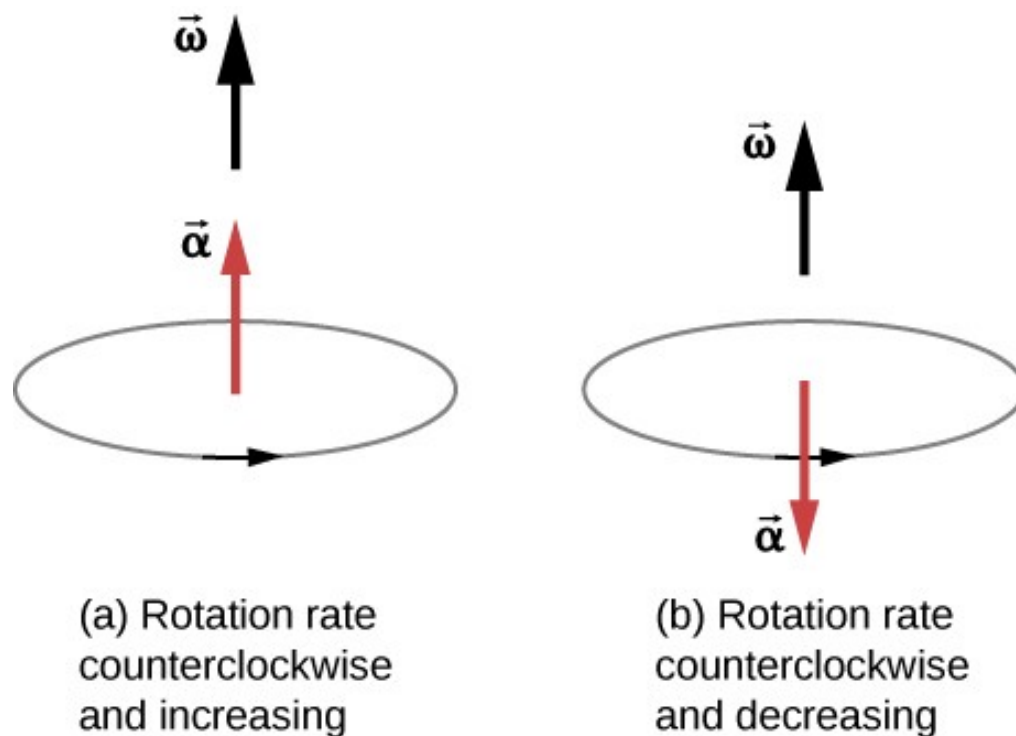


(b)

The vectors shown are the angular velocity, position, and tangential velocity.

- a) The angular velocity points in the positive z-direction, giving a counterclockwise rotation in the xy-plane.
- b) The angular velocity points in the negative z-direction, giving a clockwise rotation.

## FIGURE 10.7

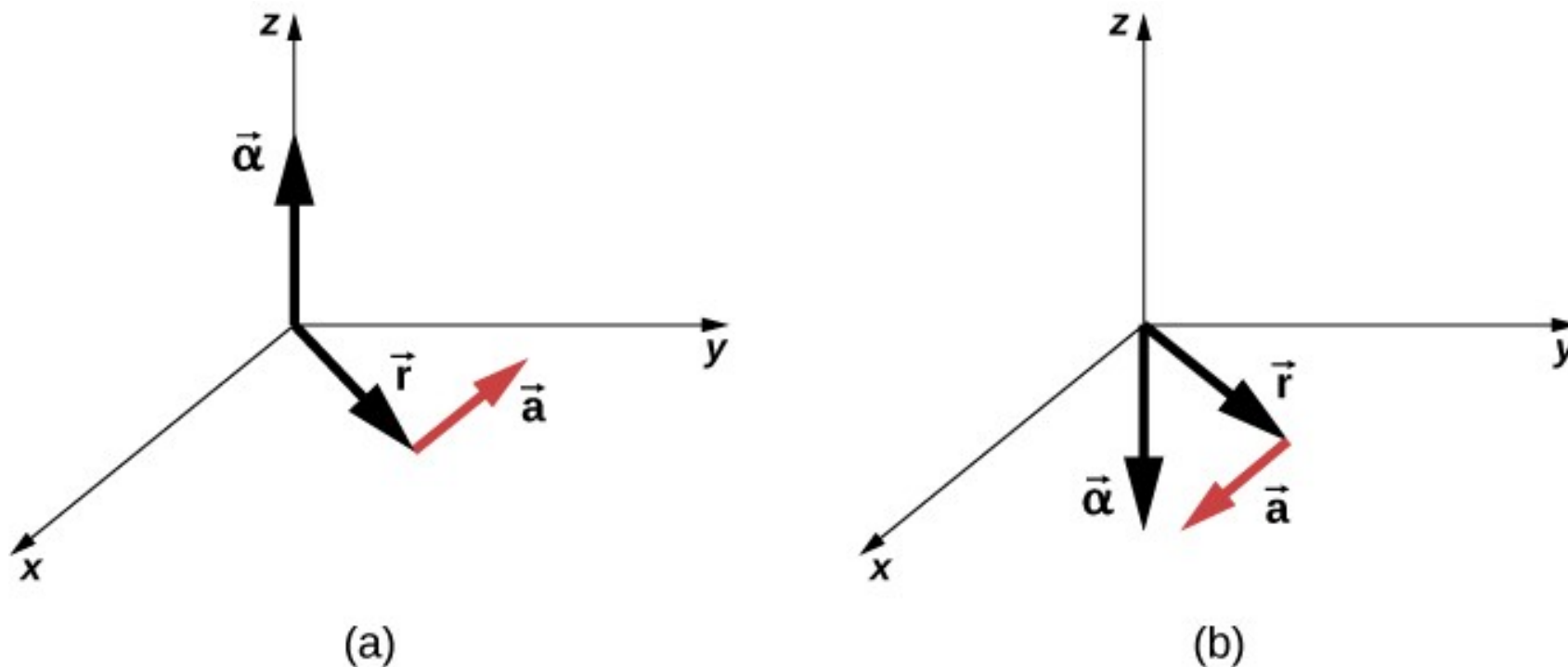


The rotation is counterclockwise in both (a) and (b) with the angular velocity in the same direction.

- a) The angular acceleration is in the same direction as the angular velocity, which increases the rotation rate.
- b) The angular acceleration is in the opposite direction to the angular velocity, which decreases the rotation rate.



## FIGURE 10.8

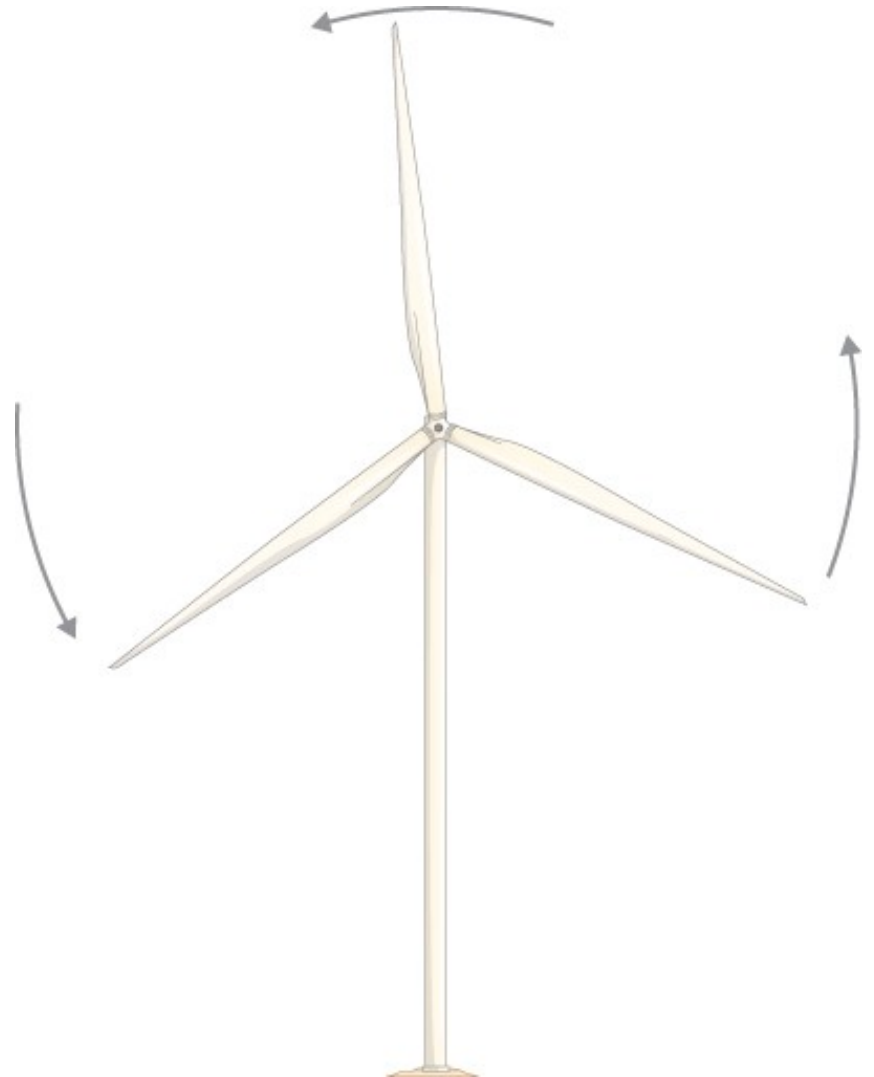


- a) The angular acceleration is the positive z-direction and produces a tangential acceleration in a counterclockwise sense.
- b) The angular acceleration is in the negative z-direction and produces a tangential acceleration in the clockwise sense.

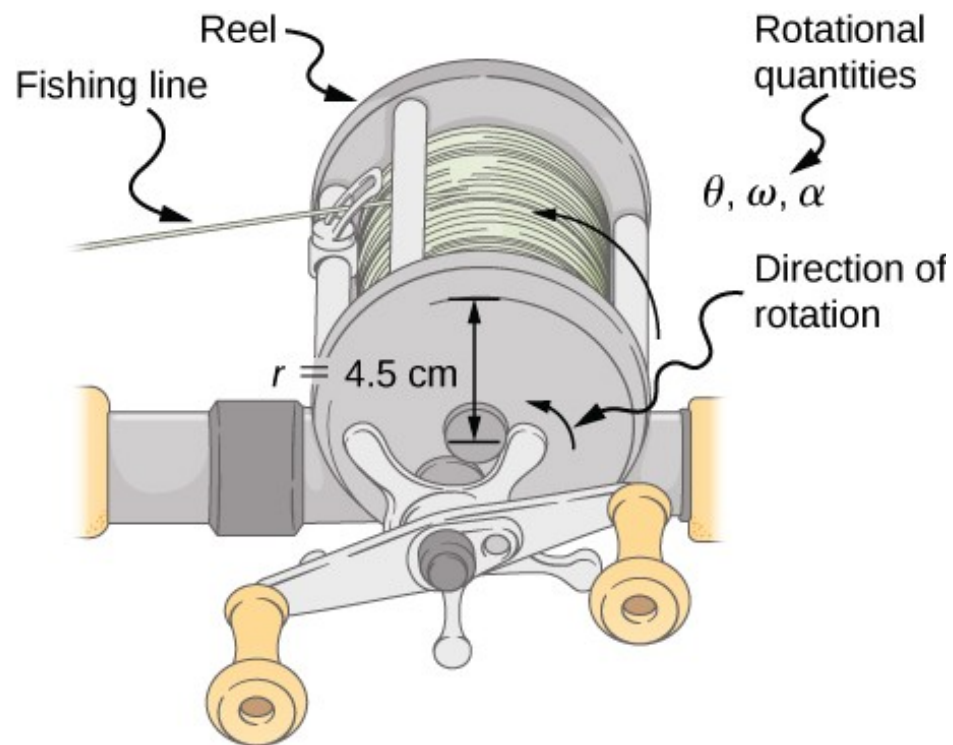
# Examples

A wind turbine that is rotating counterclockwise, as seen head on.

**FIGURE 10.10**

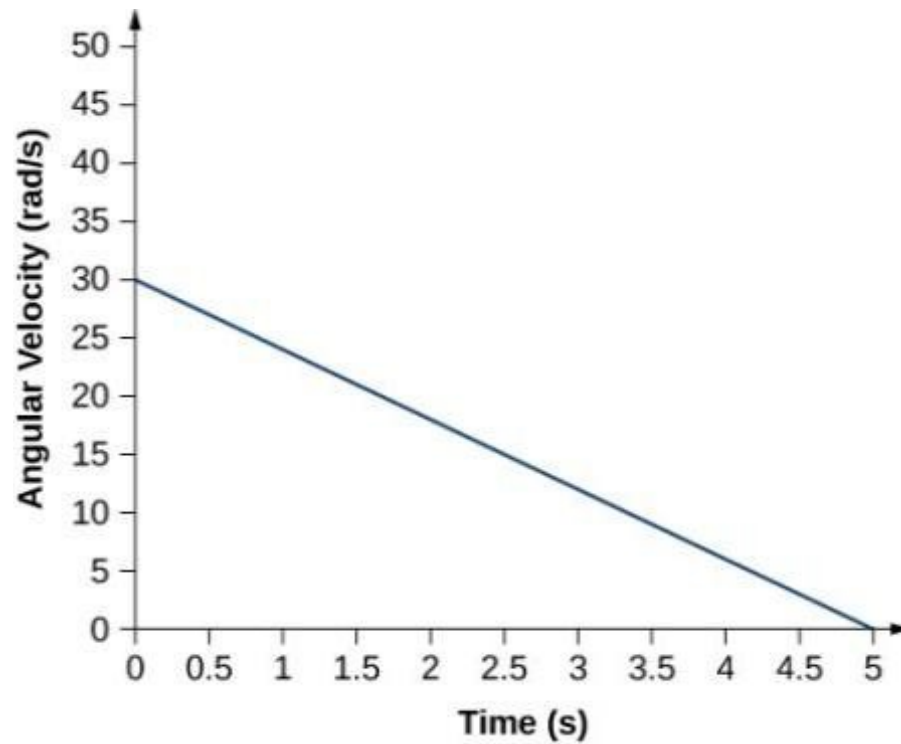


## FIGURE 10.11



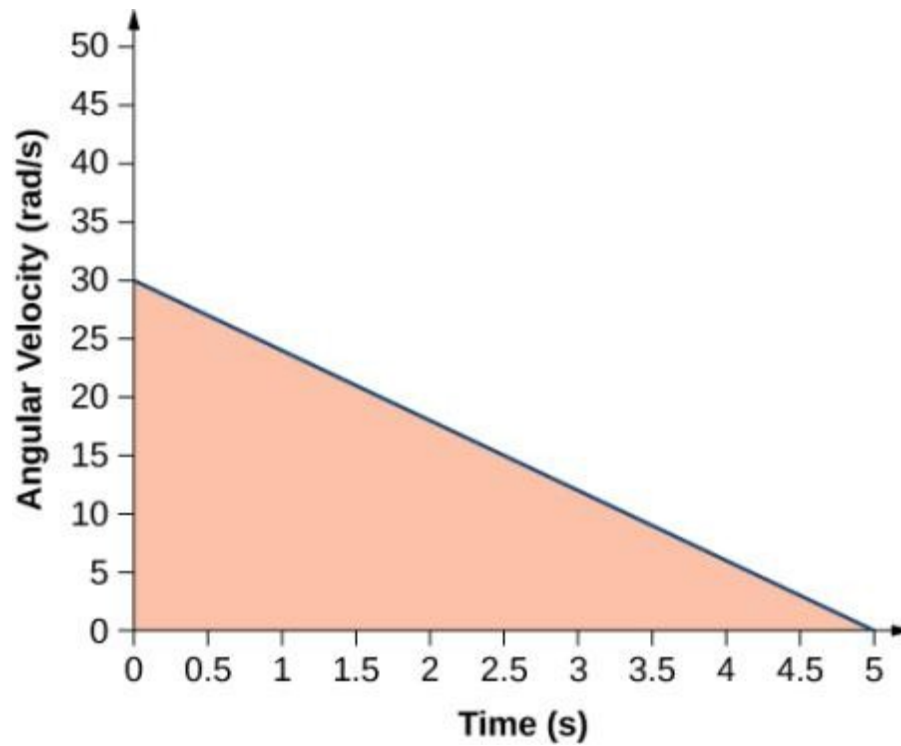
Fishing line coming off a rotating reel moves linearly.

## FIGURE 10.12



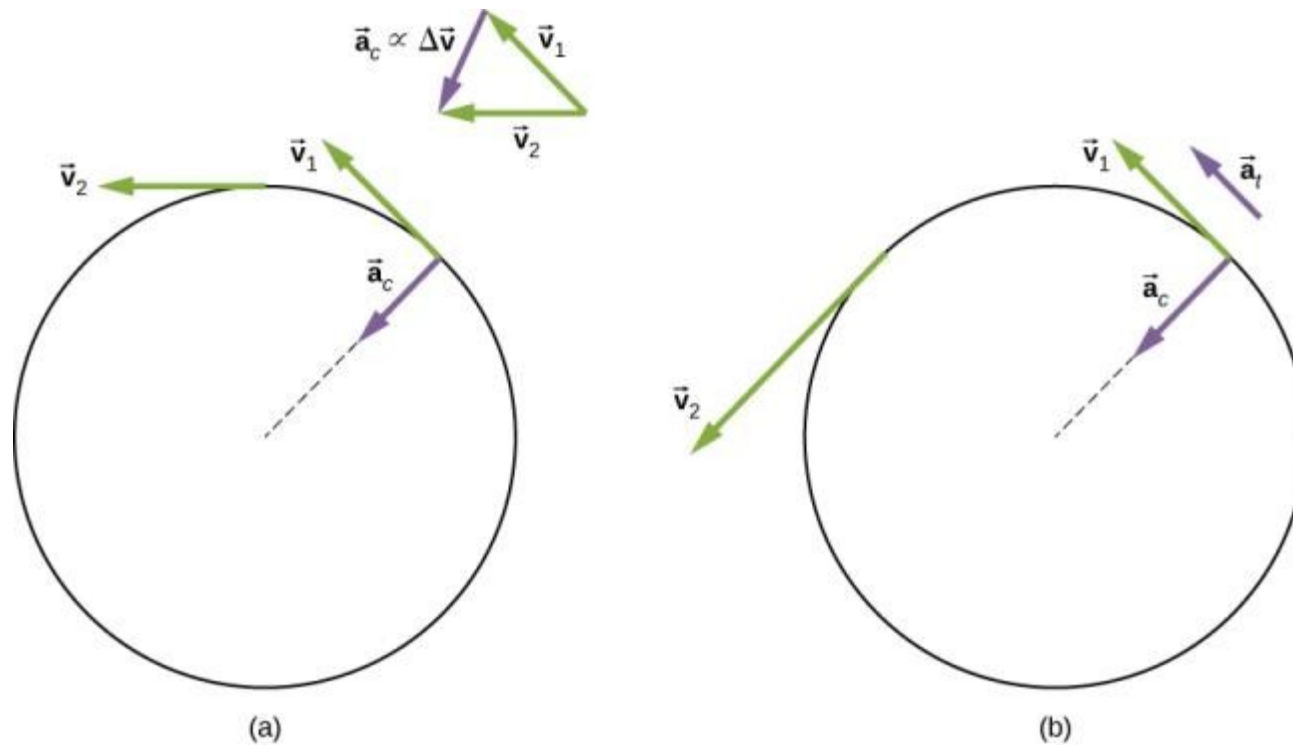
A graph of the angular velocity of a propeller versus time.

## FIGURE 10.13

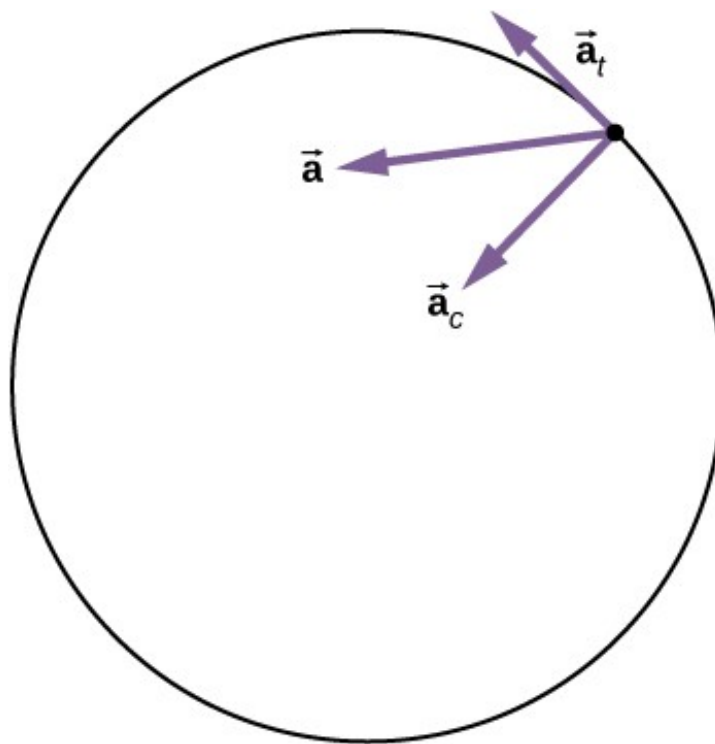


The area under the curve is the area of the right triangle.

# FIGURE 10.14



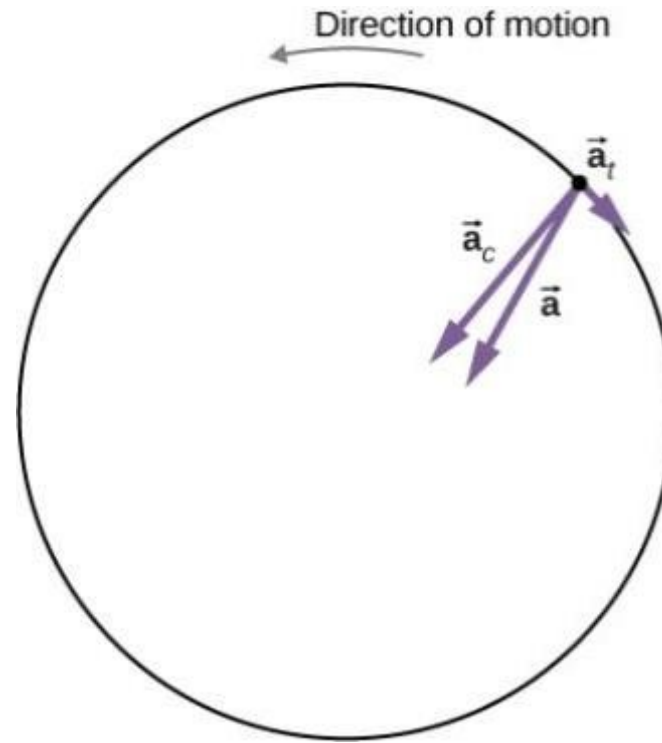
- a) Uniform circular motion: The centripetal acceleration has its vector inward toward the axis of rotation. There is no tangential acceleration.
- b) Nonuniform circular motion: An angular acceleration produces an inward centripetal acceleration that is changing in magnitude, plus a tangential acceleration .

**FIGURE 10.15**

A particle is executing circular motion and has an angular acceleration. The total linear acceleration of the particle is the vector sum of the centripetal acceleration and tangential acceleration vectors. The total linear acceleration vector is at an angle in between the centripetal and tangential accelerations.



**FIGURE 10.16**

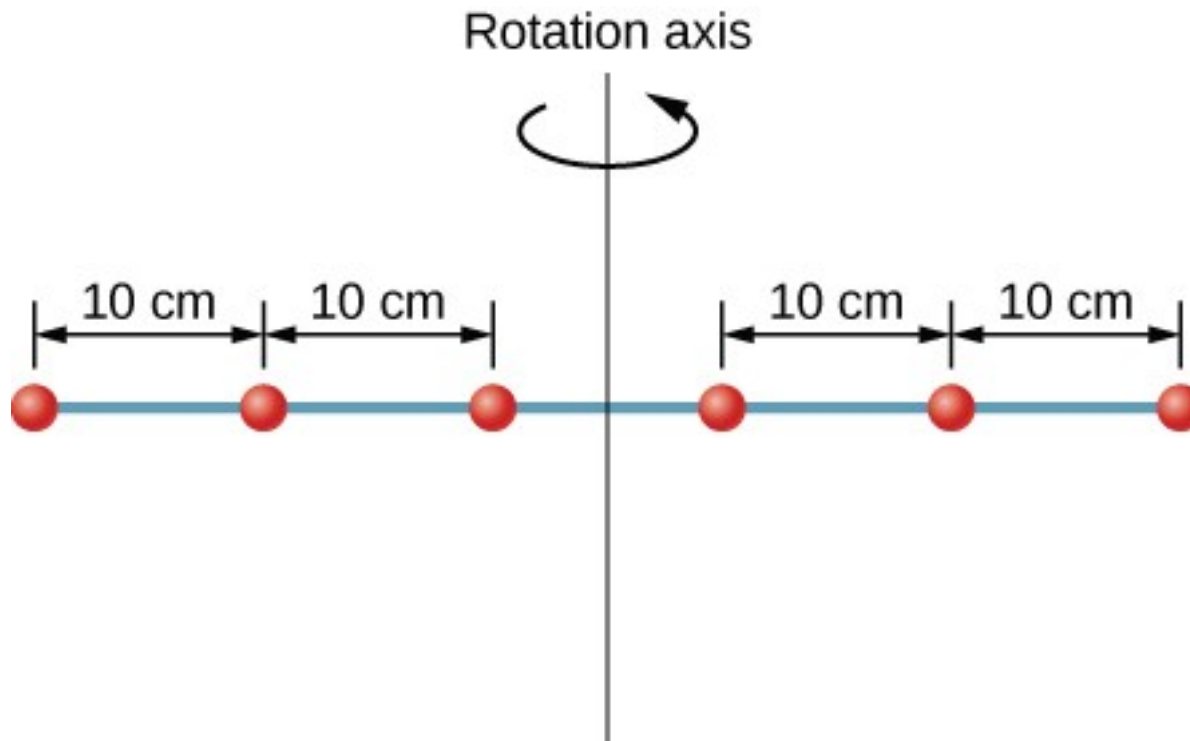


The centripetal, tangential, and total acceleration vectors. The centrifuge is slowing down, so the tangential acceleration is clockwise, opposite the direction of rotation (counterclockwise).

# Moments of inertia

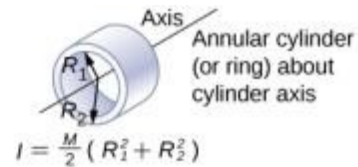
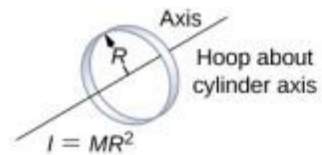
$$I = \sum m_i r_i^2 = \int r^2 dm$$

**FIGURE 10.19**

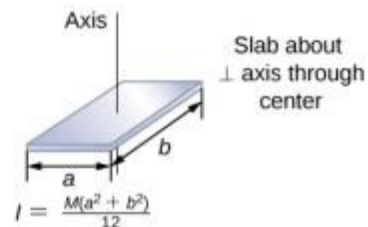
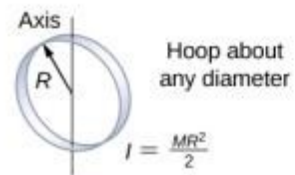
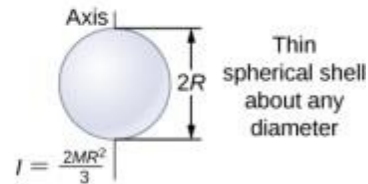
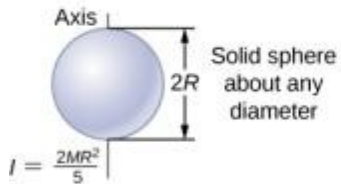
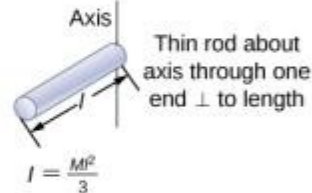
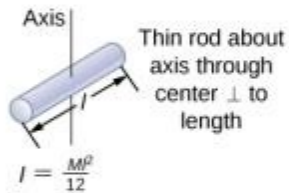
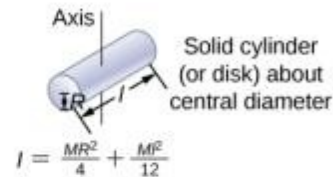
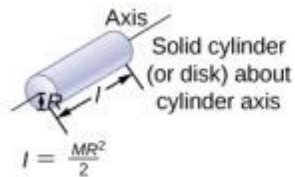


Six washers are spaced 10 cm apart on a rod of negligible mass and rotating about a vertical axis.

# FIGURE 10.20



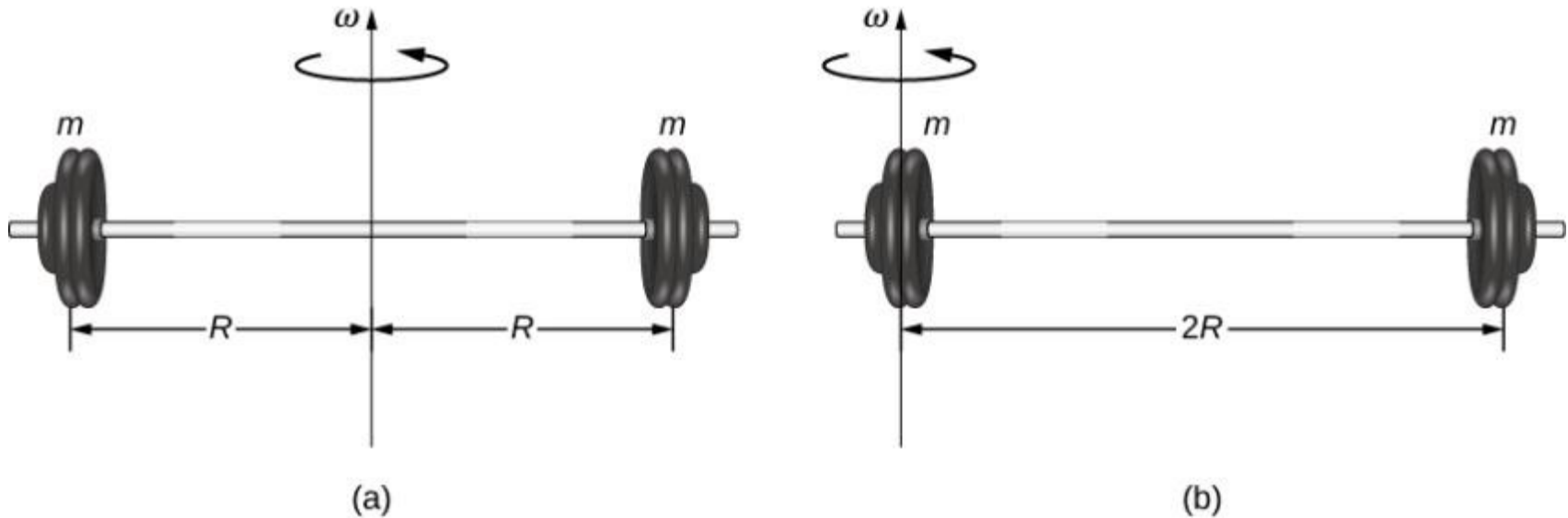
Values of rotational inertia for common shapes of objects.



# Parallel axis theorem

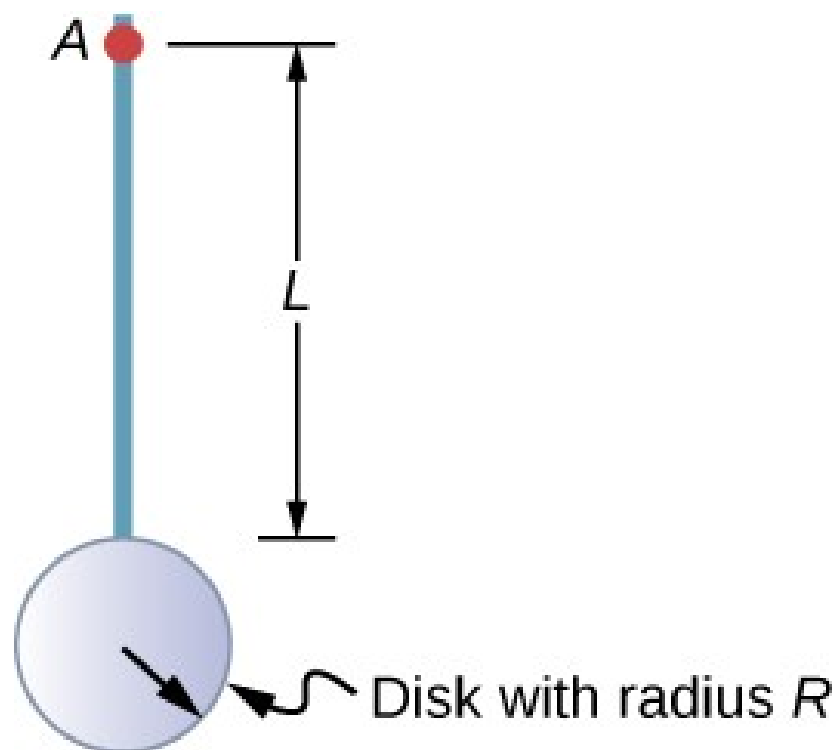
$$I = I_{cm} + m d^2$$

## FIGURE 10.23



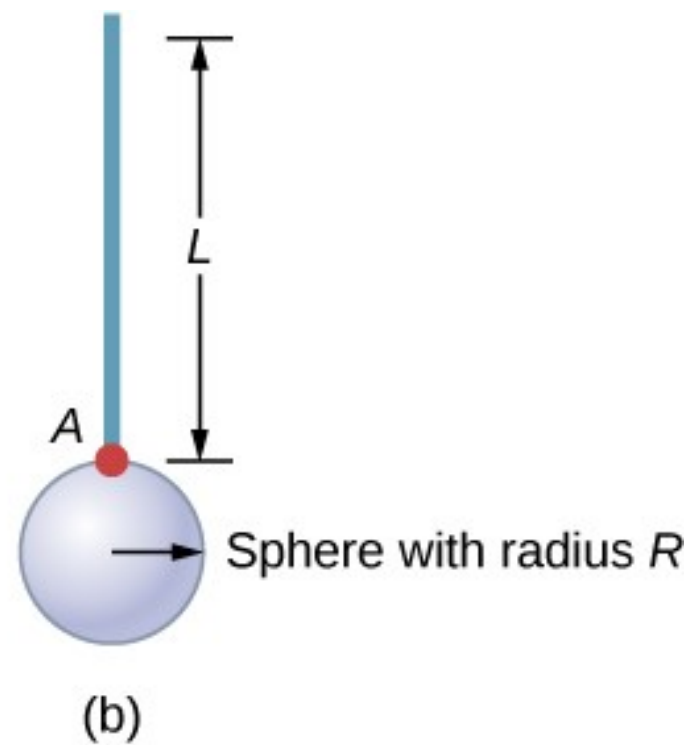
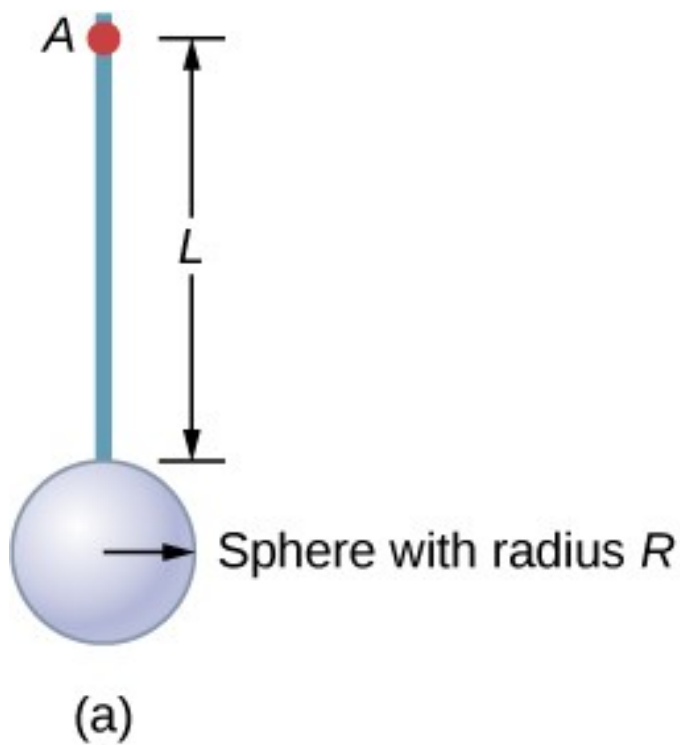
(a) A barbell with an axis of rotation through its center; (b) a barbell with an axis of rotation through one end.

**FIGURE 10.28**



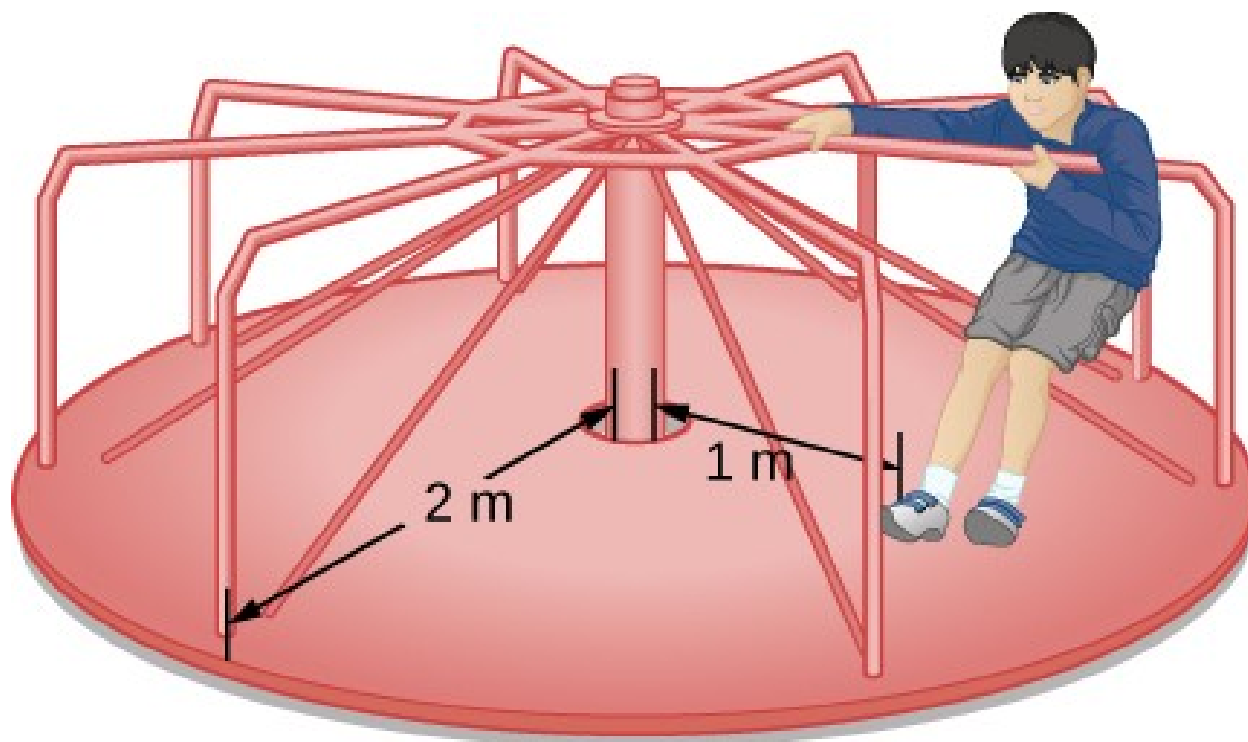
Compound object consisting of a disk at the end of a rod. The axis of rotation is located at  $A$ .

## EXAMPLE 10.12



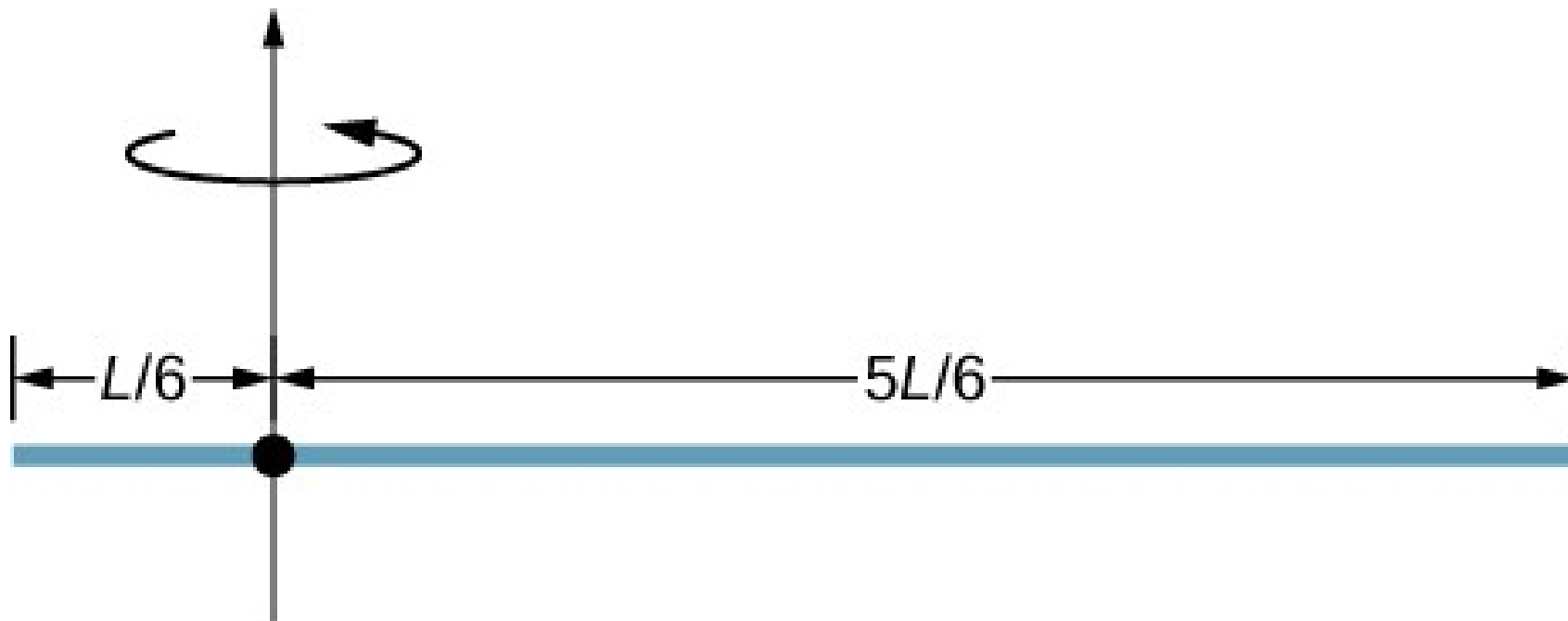


**FIGURE 10.29**



Calculating the moment of inertia for a child on a merry-go-round.

## EXERCISE 65

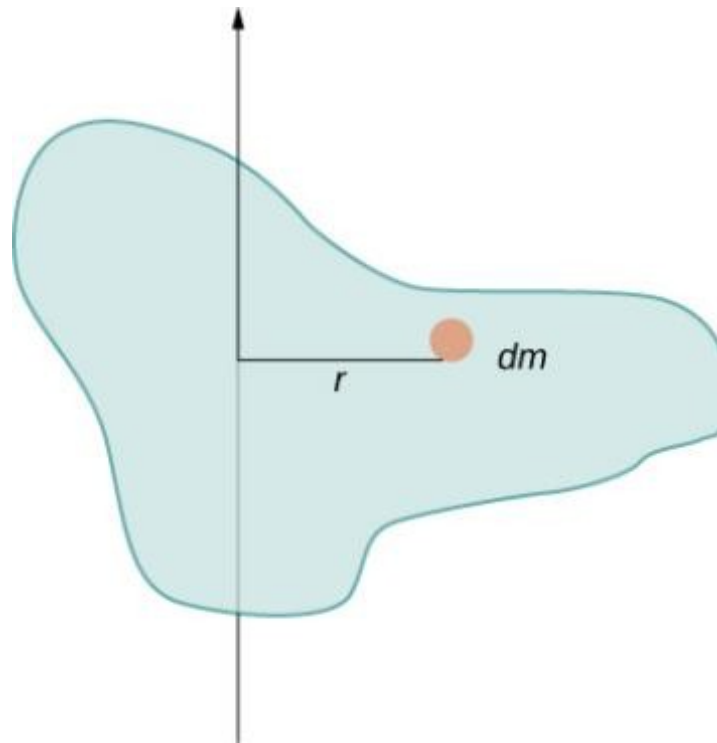


Using the parallel axis theorem, what is the moment of inertia of the rod of mass  $m$  about the axis shown below?

# Moments of inertia - integrating

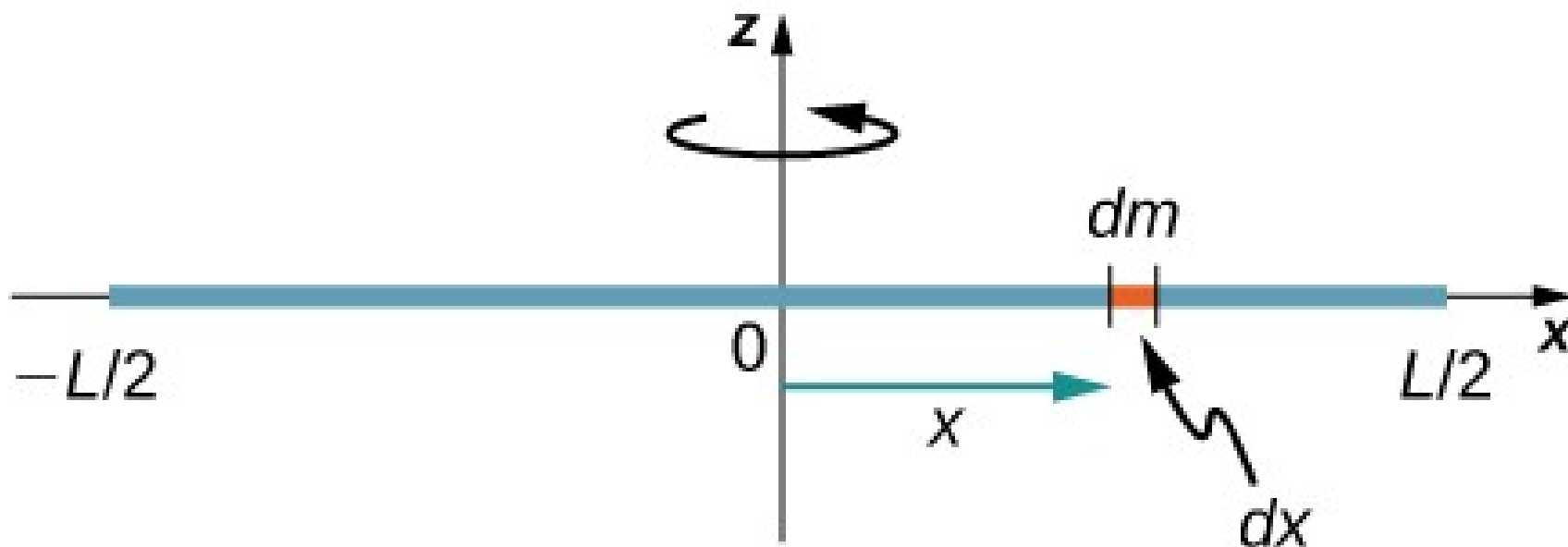
$$I = \int r^2 dm$$

## FIGURE 10.24



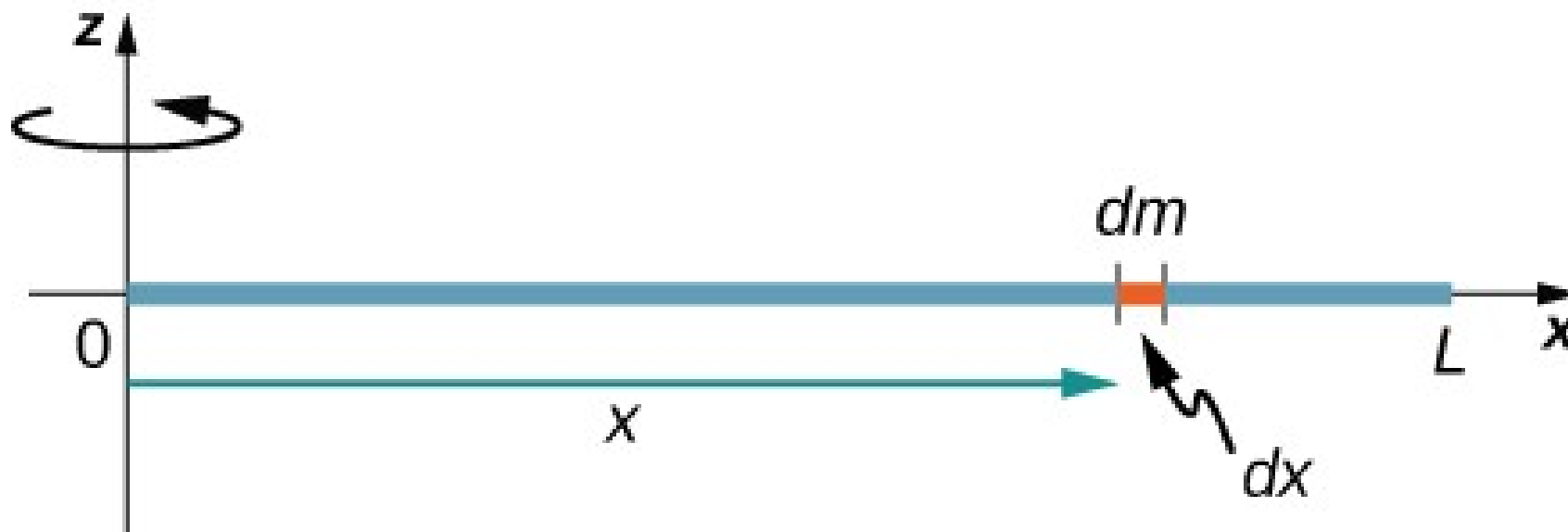
Using an infinitesimally small piece of mass to calculate the contribution to the total moment of inertia.

**FIGURE 10.25**



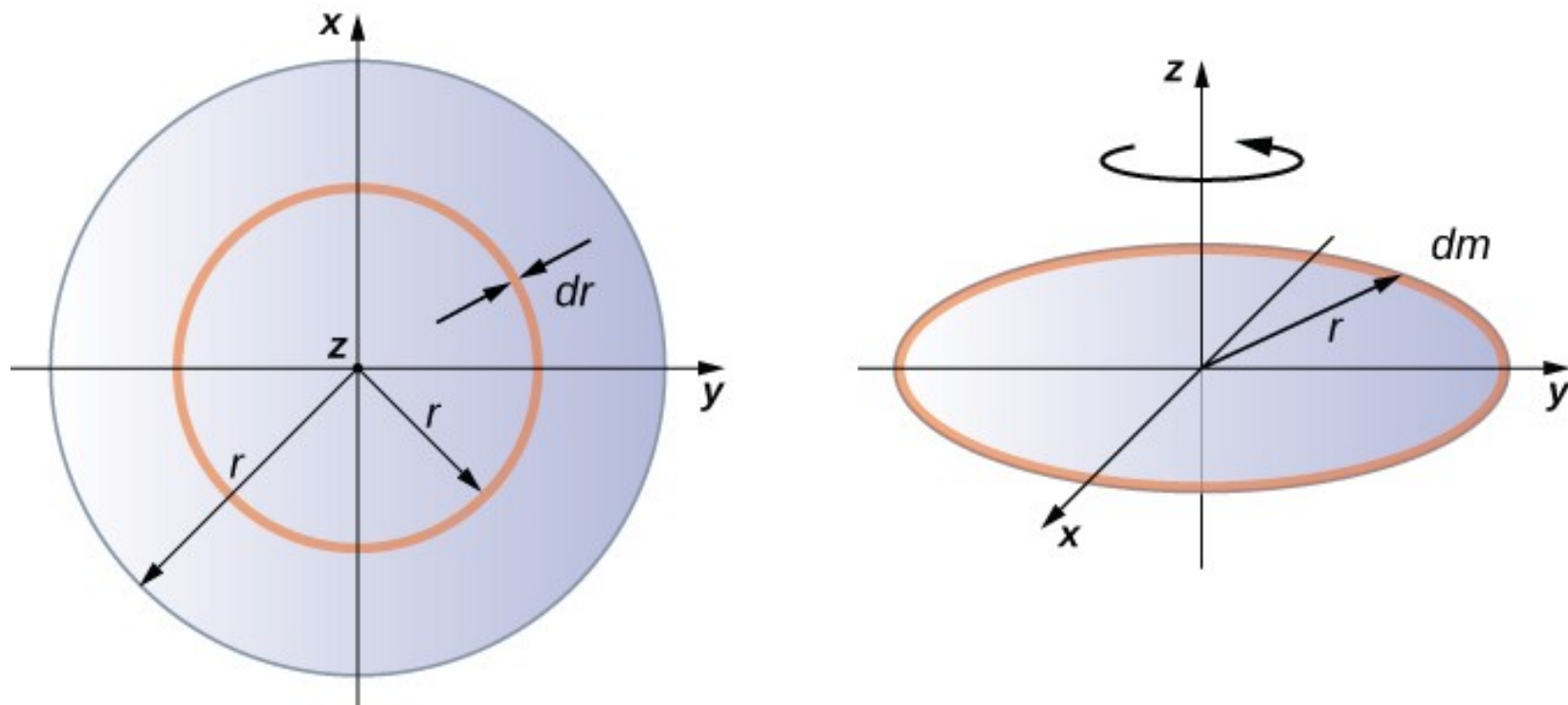
Calculation of the moment of inertia  $I$  for a uniform thin rod about an axis through the center of the rod.

**FIGURE 10.26**



Calculation of the moment of inertia  $I$  for a uniform thin rod about an axis through the end of the rod.

## FIGURE 10.27



Calculating the moment of inertia for a thin disk about an axis through its center.

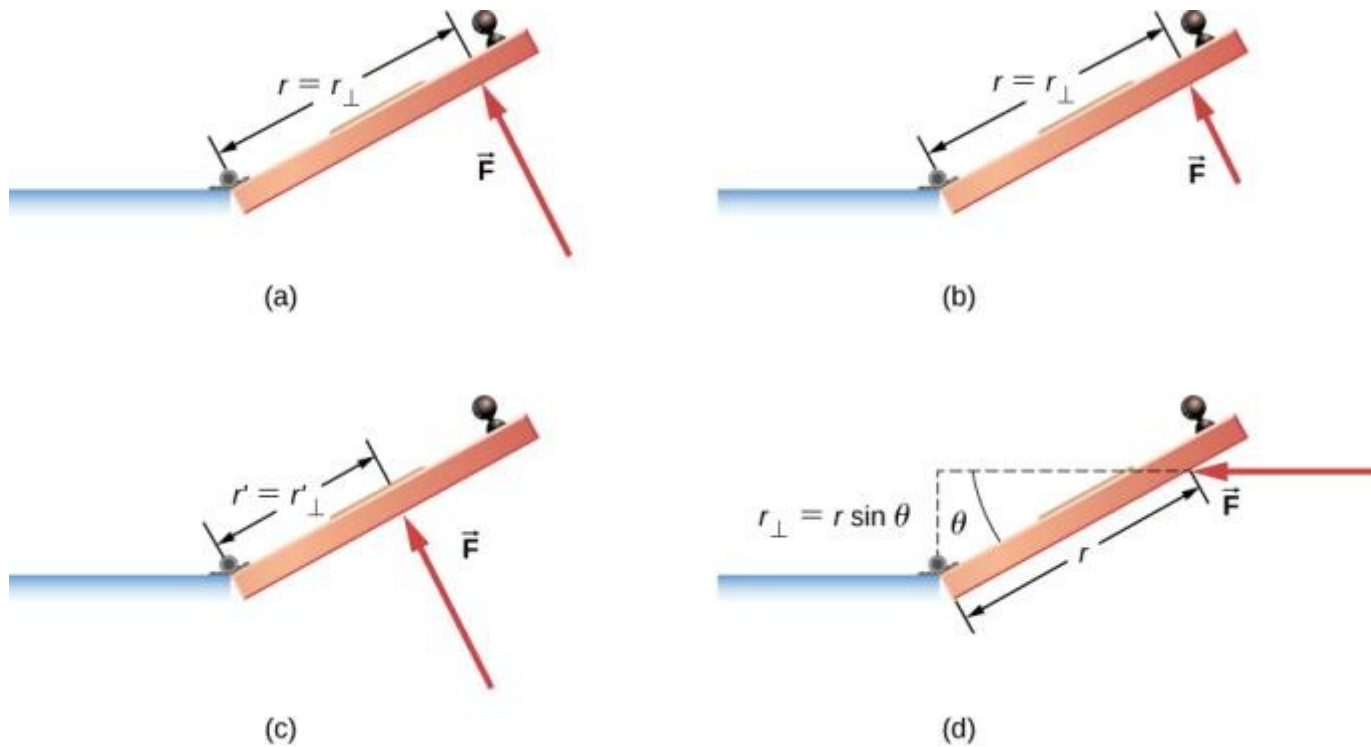
# Torque

$$\vec{\tau} = \vec{r} \times \vec{F}$$

$$\vec{\tau} = I \vec{\alpha}$$



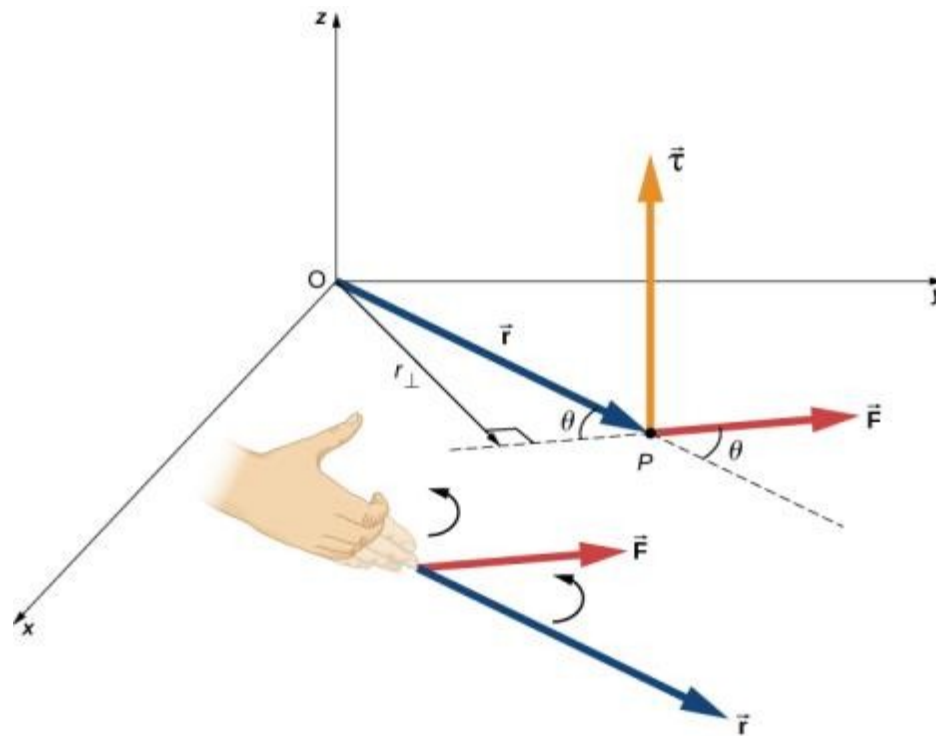
## FIGURE 10.31



Torque is the turning or twisting effectiveness of a force, illustrated here for door rotation on its hinges (as viewed from overhead). Torque has both magnitude and direction.

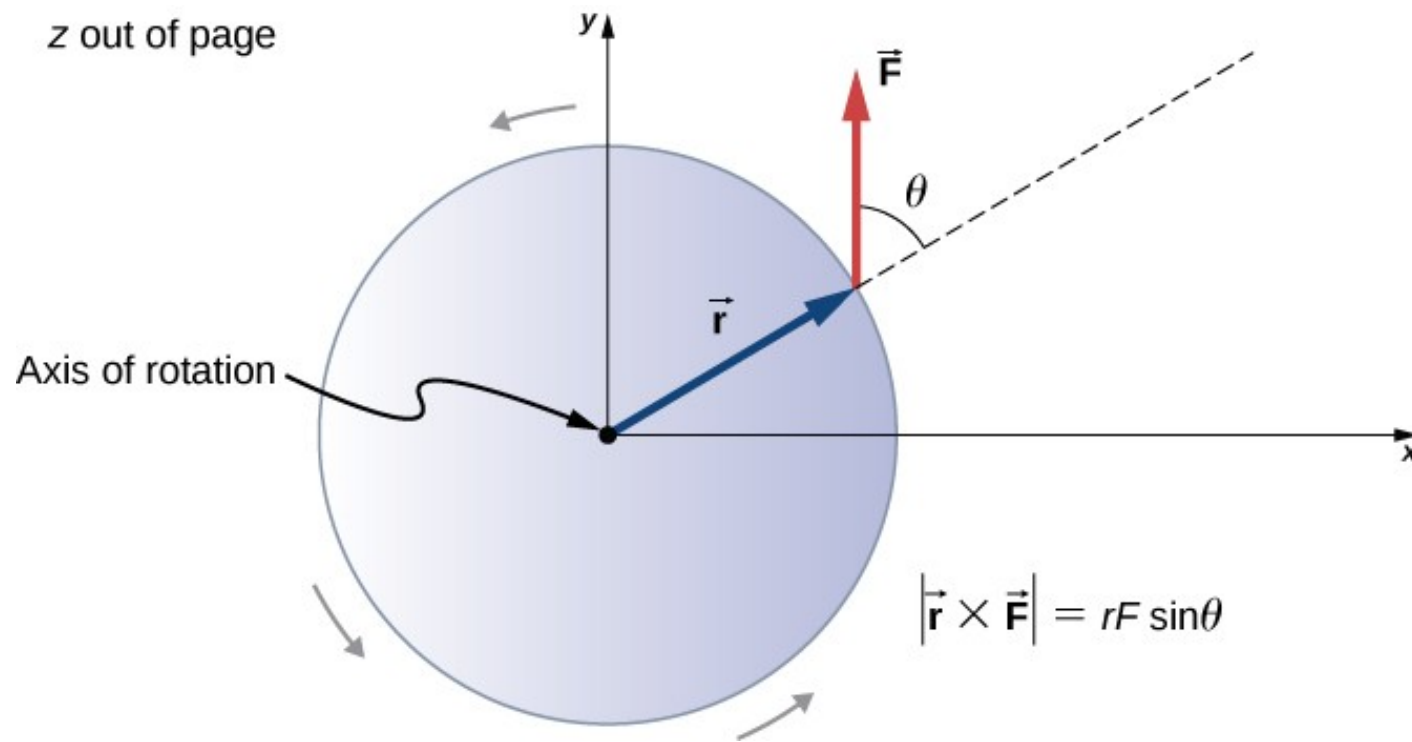
- a) A counterclockwise torque is produced by a force acting at a distance  $r$  from the hinges (the pivot point).
- b) A smaller counterclockwise torque is produced when a smaller force  $F'$  acts at the same distance  $r$  from the hinges.
- c) The same force as in (a) produces a smaller counterclockwise torque when applied at a smaller distance from the hinges.
- d) A smaller counterclockwise torque is produced by the same magnitude force as (a) acting at the same distance as (a) but at an angle that is less than  $90^\circ$ .

## FIGURE 10.32



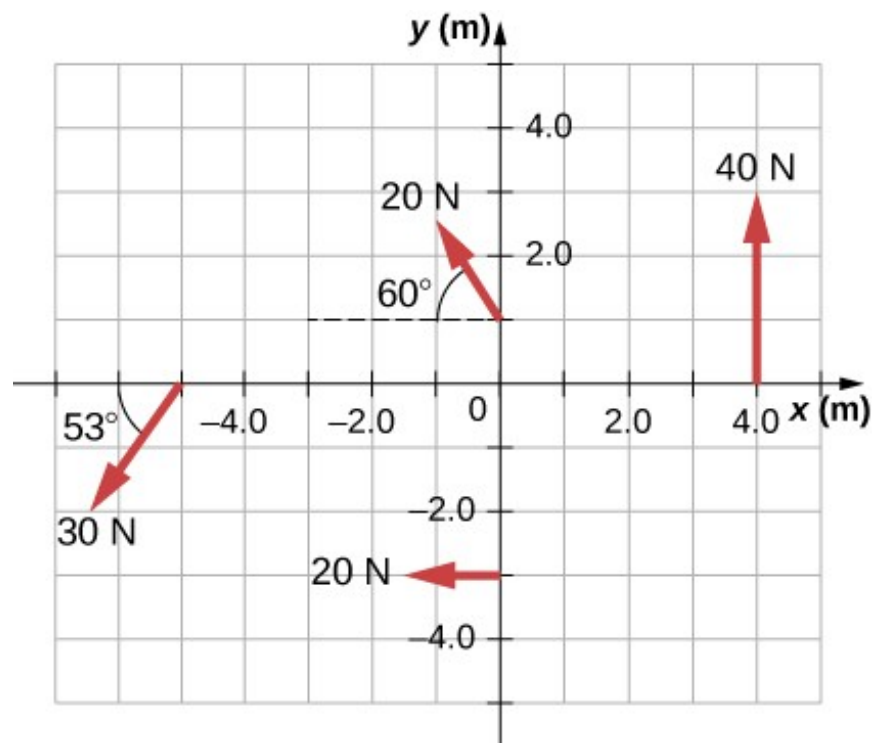
The torque is perpendicular to the plane defined by  $\vec{r}$  and  $\vec{F}$  and its direction is determined by the right-hand rule.

## FIGURE 10.33



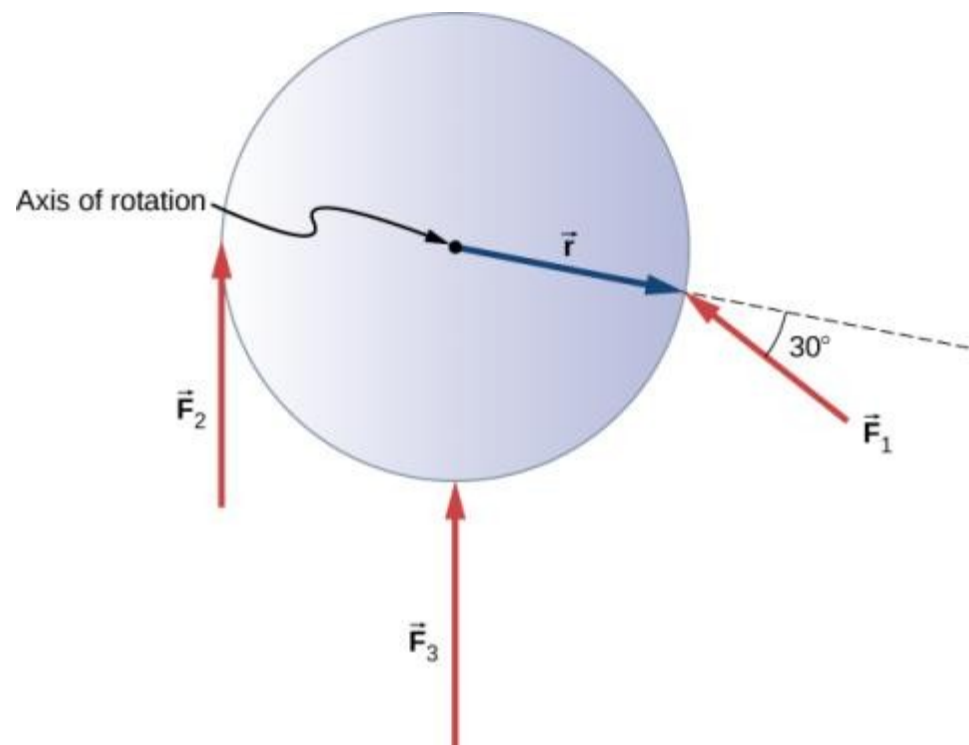
A disk is free to rotate about its axis through the center. The magnitude of the torque on the disk is  $rF \sin\theta$ . When  $\theta = 0^\circ$ , the torque is zero and the disk does not rotate. When  $\theta = 90^\circ$ , the torque is maximum and the disk rotates with maximum angular acceleration.

## FIGURE 10.34



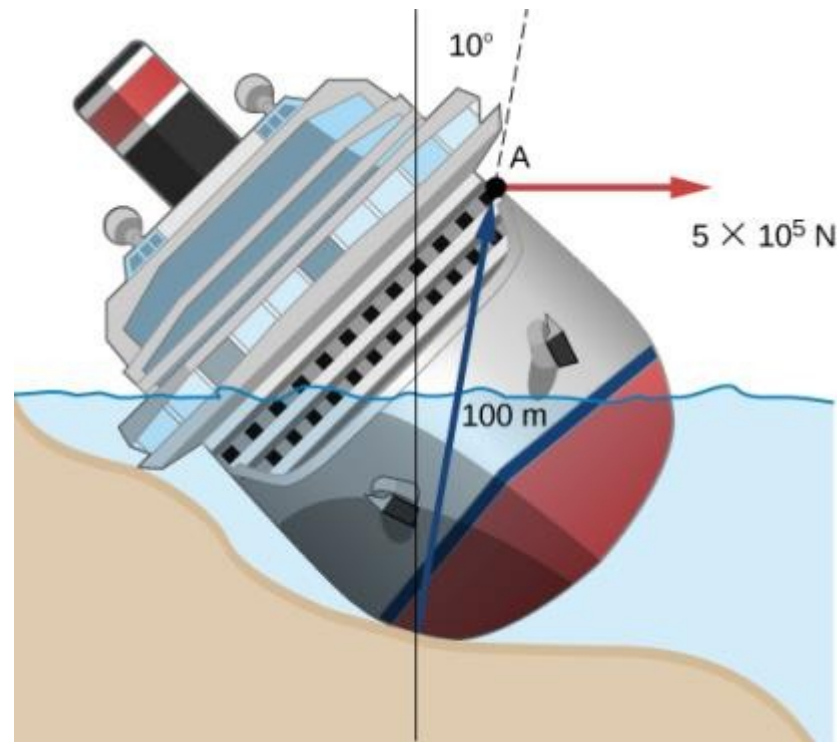
Four forces producing torques.

## FIGURE 10.35



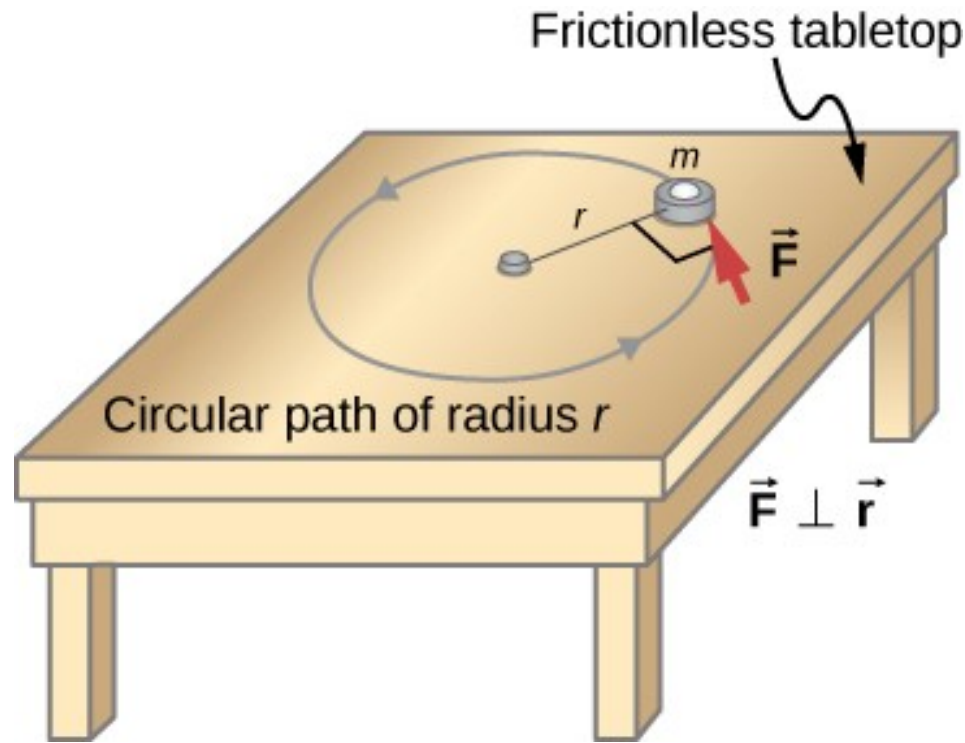
Three forces acting on a flywheel.

**FIGURE 10.36**



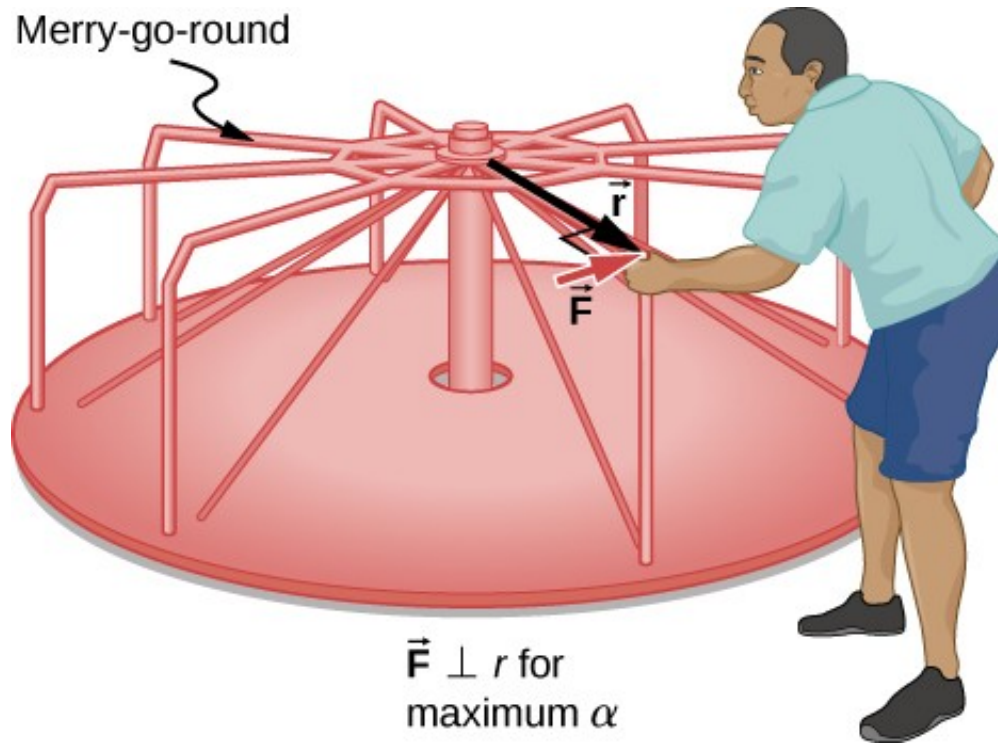
A ship runs aground and tilts, requiring torque to be applied to return the vessel to an upright position.

## FIGURE 10.37



An object is supported by a horizontal frictionless table and is attached to a pivot point by a cord that supplies centripetal force. A force is applied to the object perpendicular to the radius  $r$ , causing it to accelerate about the pivot point. The force is perpendicular to  $r$ .

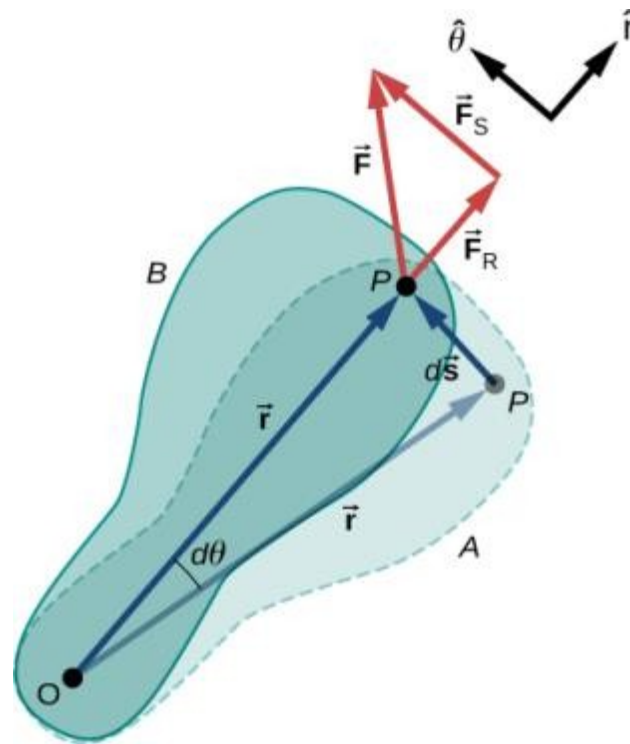
## FIGURE 10.38



A father pushes a playground merry-go-round at its edge and perpendicular to its radius to achieve maximum torque.



## FIGURE 10.39



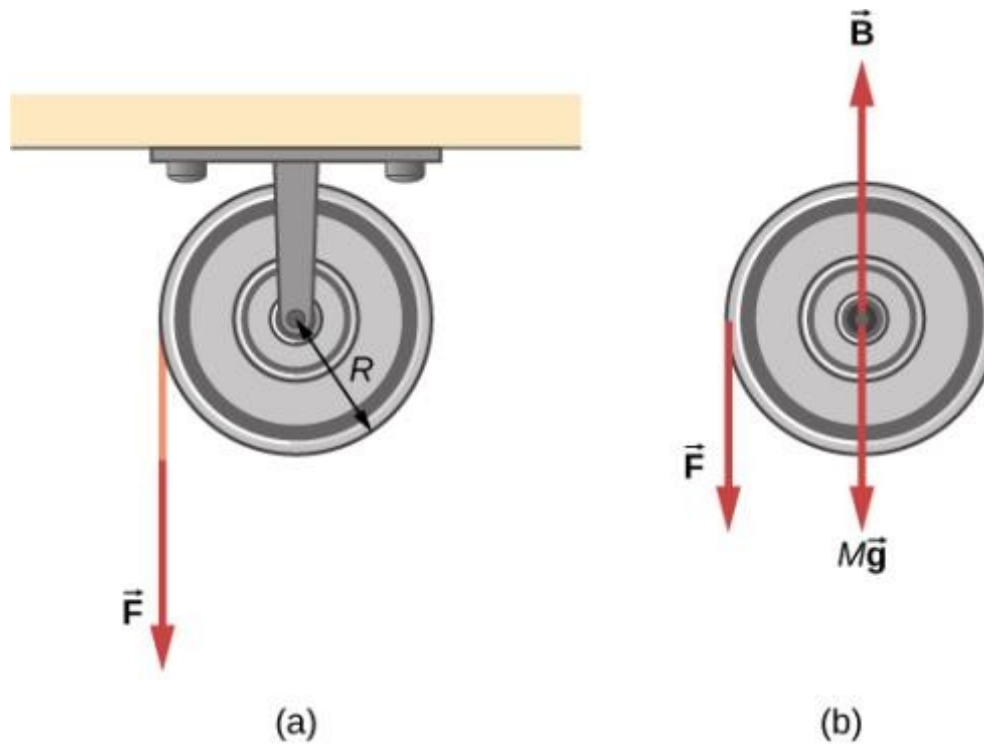
A rigid body rotates through an angle  $d\theta$  from  $A$  to  $B$  by the action of an external force applied to point  $P$ .

# Kinematic problems

$$\vec{\tau} = \vec{r} \times \vec{F}$$

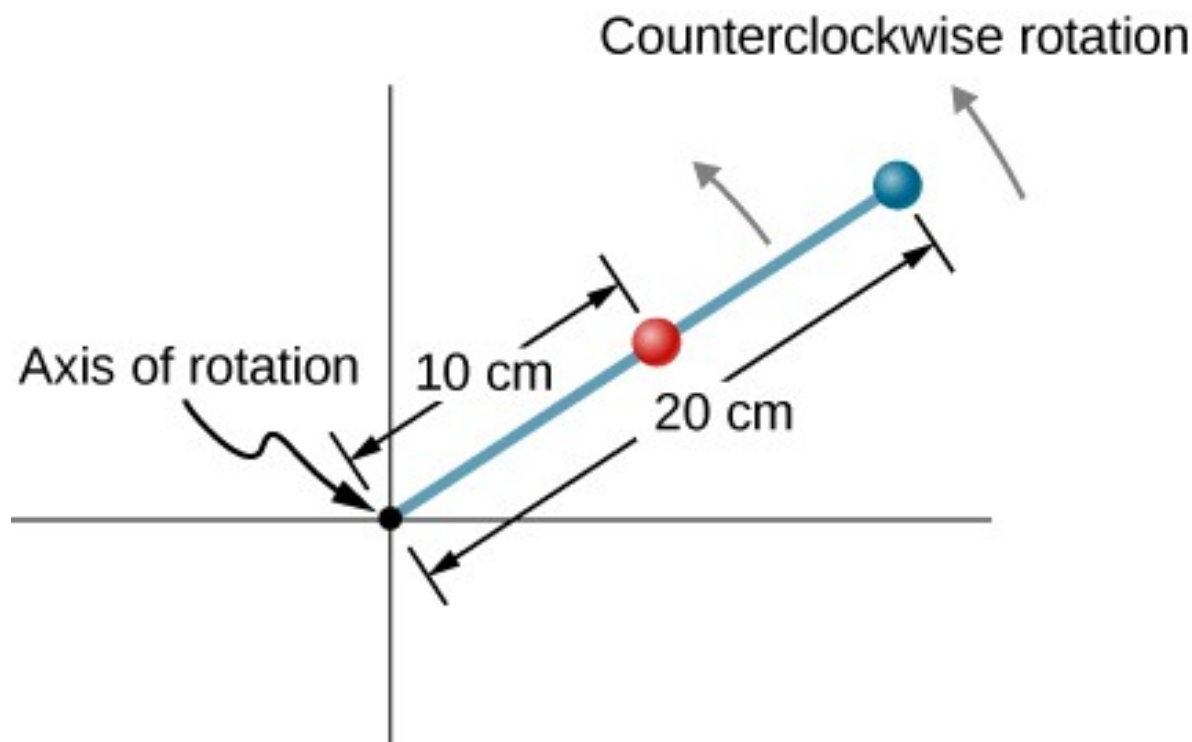
$$\vec{\tau} = I \vec{\alpha}$$

## FIGURE 10.40



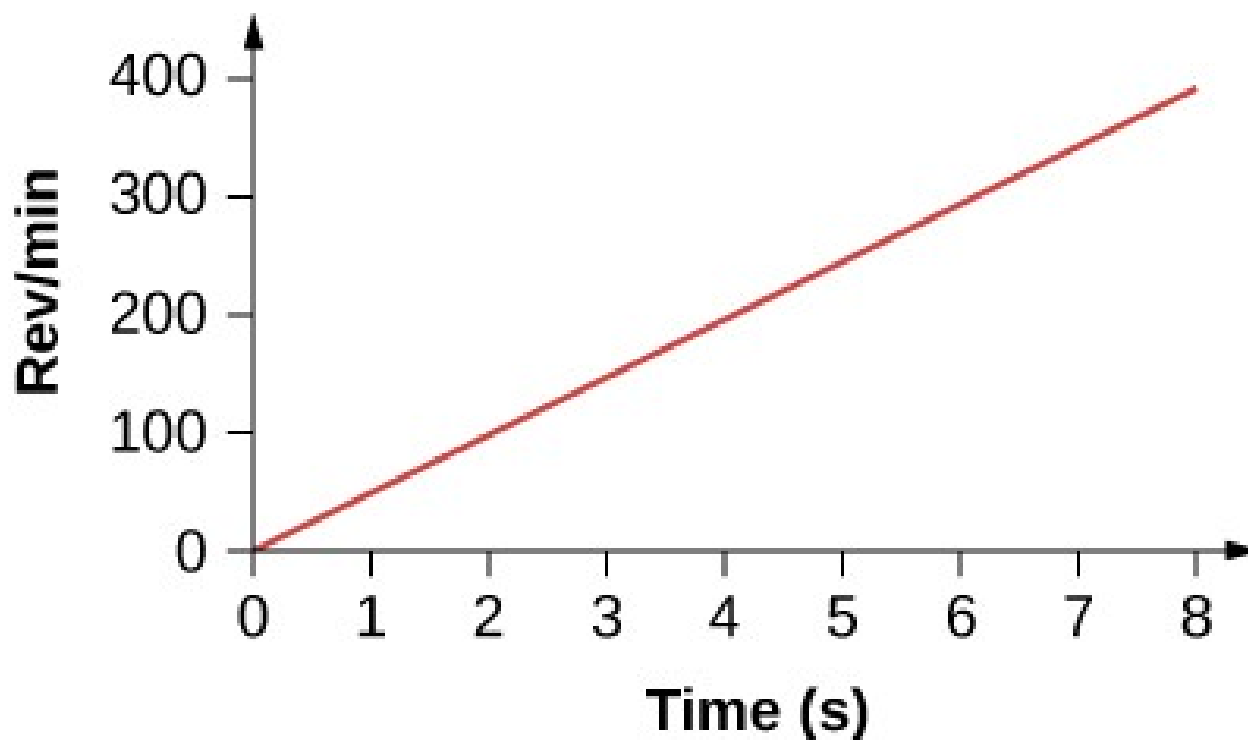
- a) A string is wrapped around a pulley of radius  $R$ .
- b) The free-body diagram.

## EXERCISE 36



The angular position of a rod varies as [equation]. The rod has two beads on it as shown in the following figure, one at 10 cm from the rotation axis and the other at 20 cm from the rotation axis. (a) What is the instantaneous angular velocity of the rod at [time]? (b) What is the angular acceleration of the rod? (c) What are the tangential speeds of the beads at [time]? (d) What are the tangential accelerations of the beads at [time]? (e) What are the centripetal accelerations of the beads at [time]?

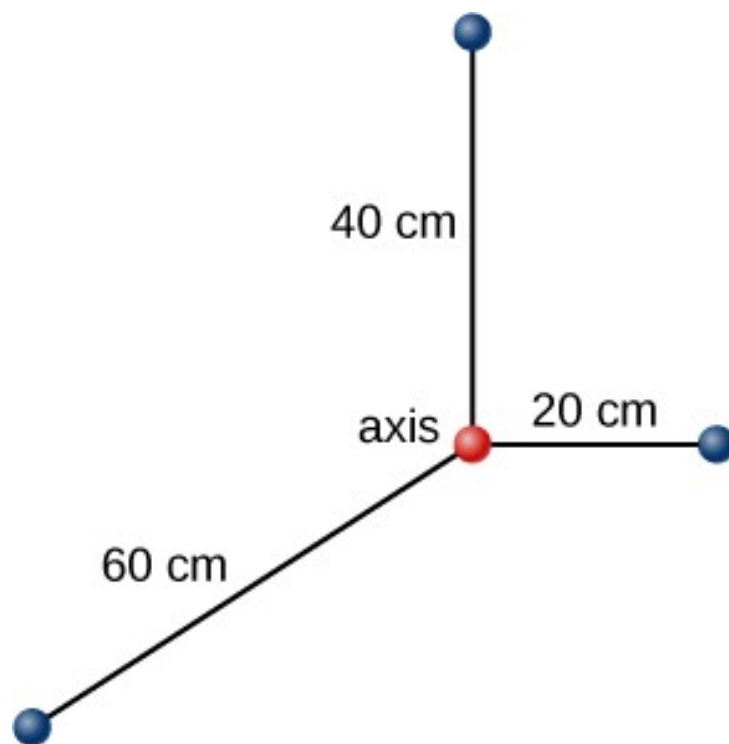
## EXERCISE 44



The angular velocity vs. time for a fan on a hovercraft is shown below.  
(a) What is the angle through which the fan blades rotate in the first 8 seconds? (b) Verify your result using the kinematic equations.

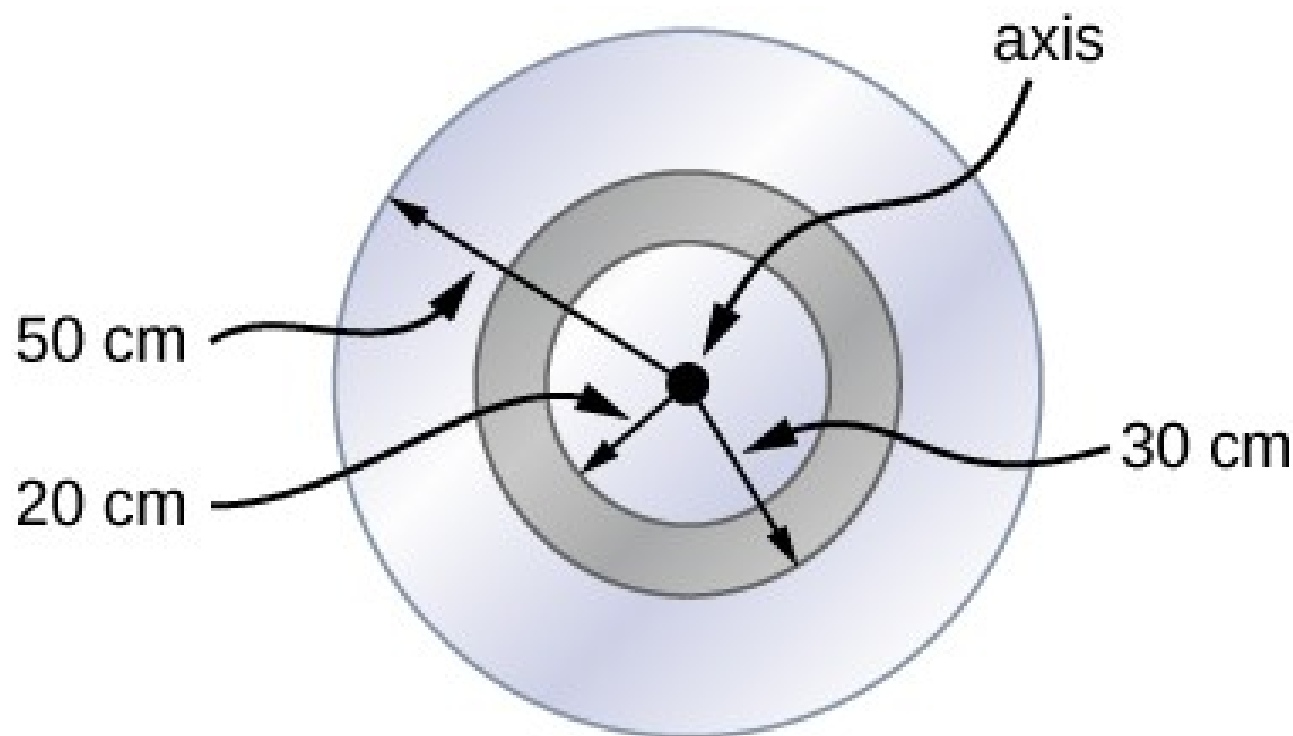
# Examples

## EXERCISE 54



A system of point particles is shown in the following figure. Each particle has mass  $m$  and they all lie in the same plane. (a) What is the moment of inertia of the system about the given axis? (b) If the system rotates at  $[\omega]$ , what is its rotational kinetic energy?

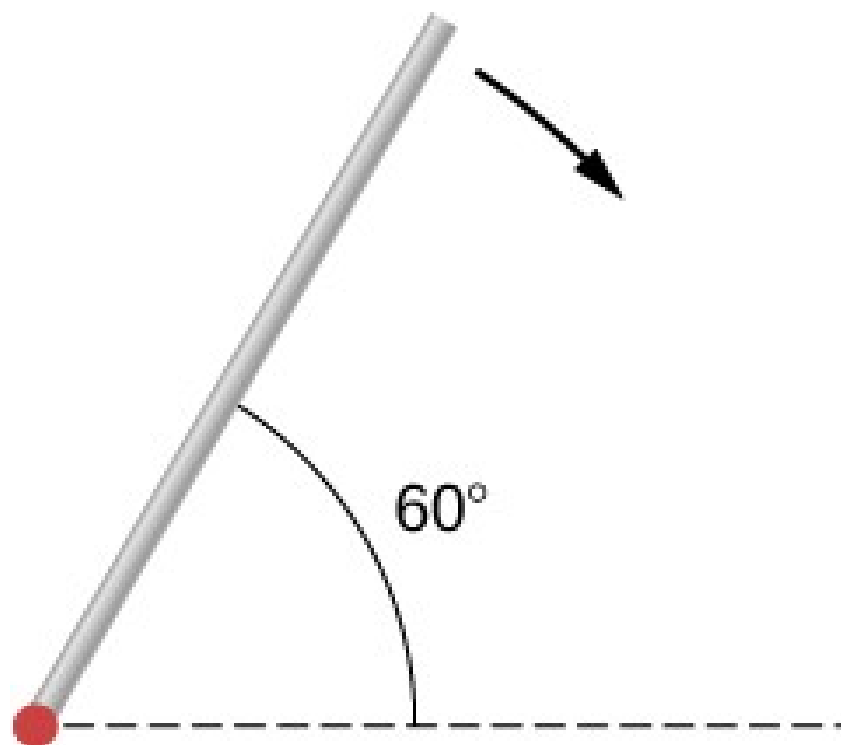
## EXERCISE 63



A system consists of a disk of mass 2.0 kg and radius 50 cm upon which is mounted an annular cylinder of mass 1.0 kg with inner radius 20 cm and outer radius 30 cm (see below). The system rotates about an axis through the center of the disk and annular cylinder at 10 rev/s. (a) What is the moment of inertia of the system? (b) What is its rotational kinetic energy?

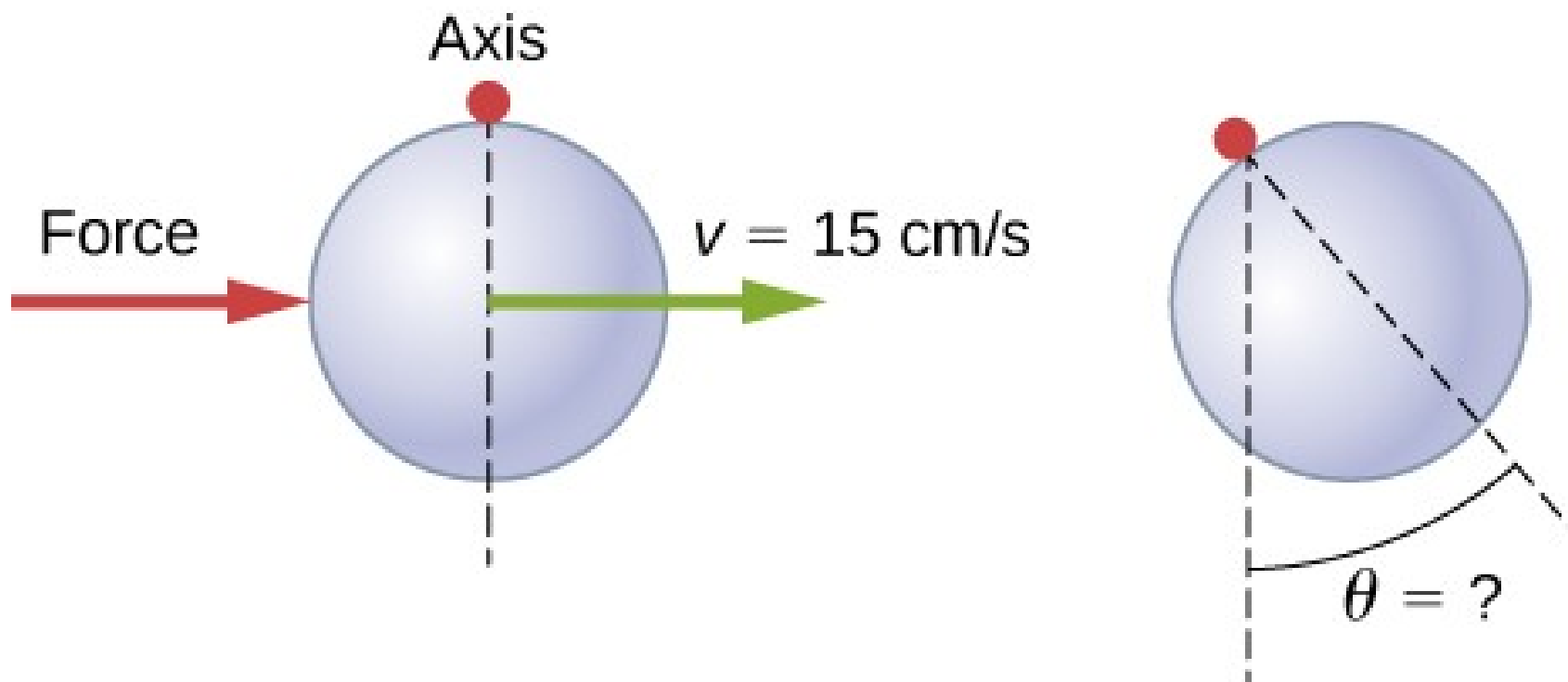


## EXERCISE 67



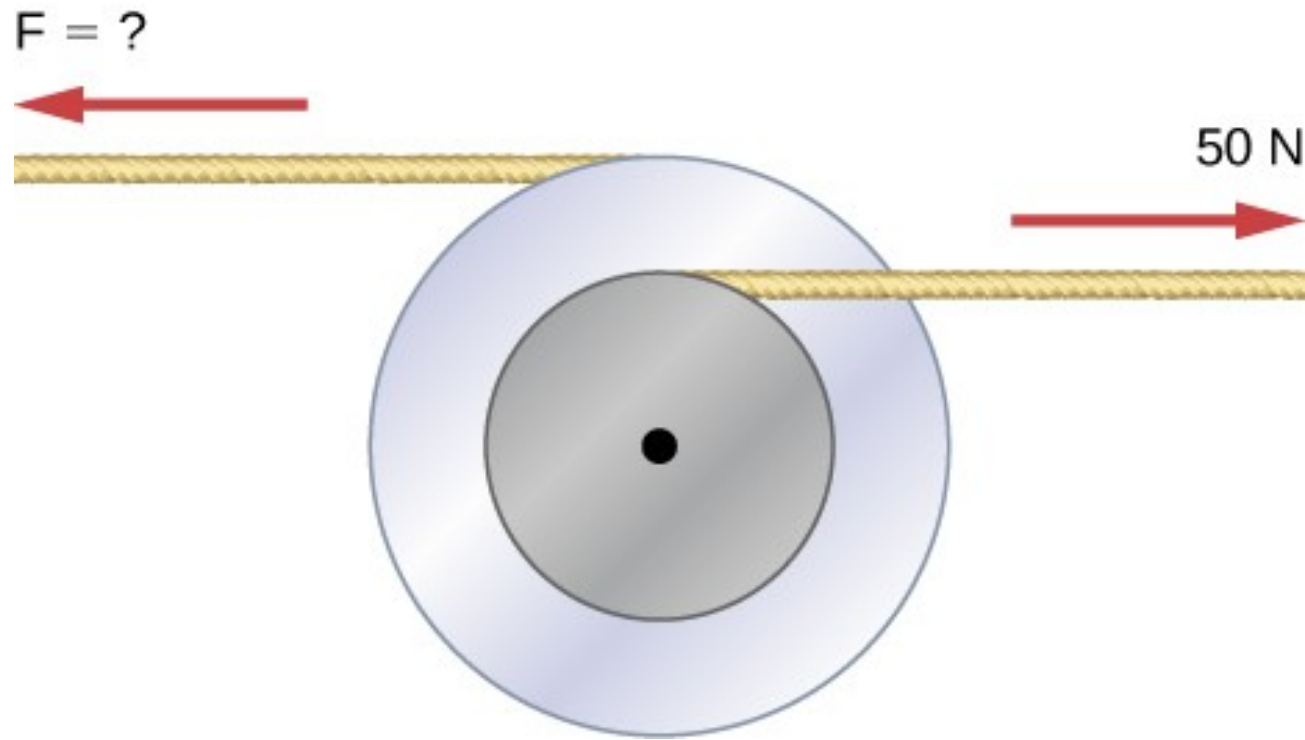
A uniform rod of mass 1.0 kg and length 2.0 m is free to rotate about one end (see the following figure). If the rod is released from rest at an angle of [angle] with respect to the horizontal, what is the speed of the tip of the rod as it passes the horizontal position?

## EXERCISE 69



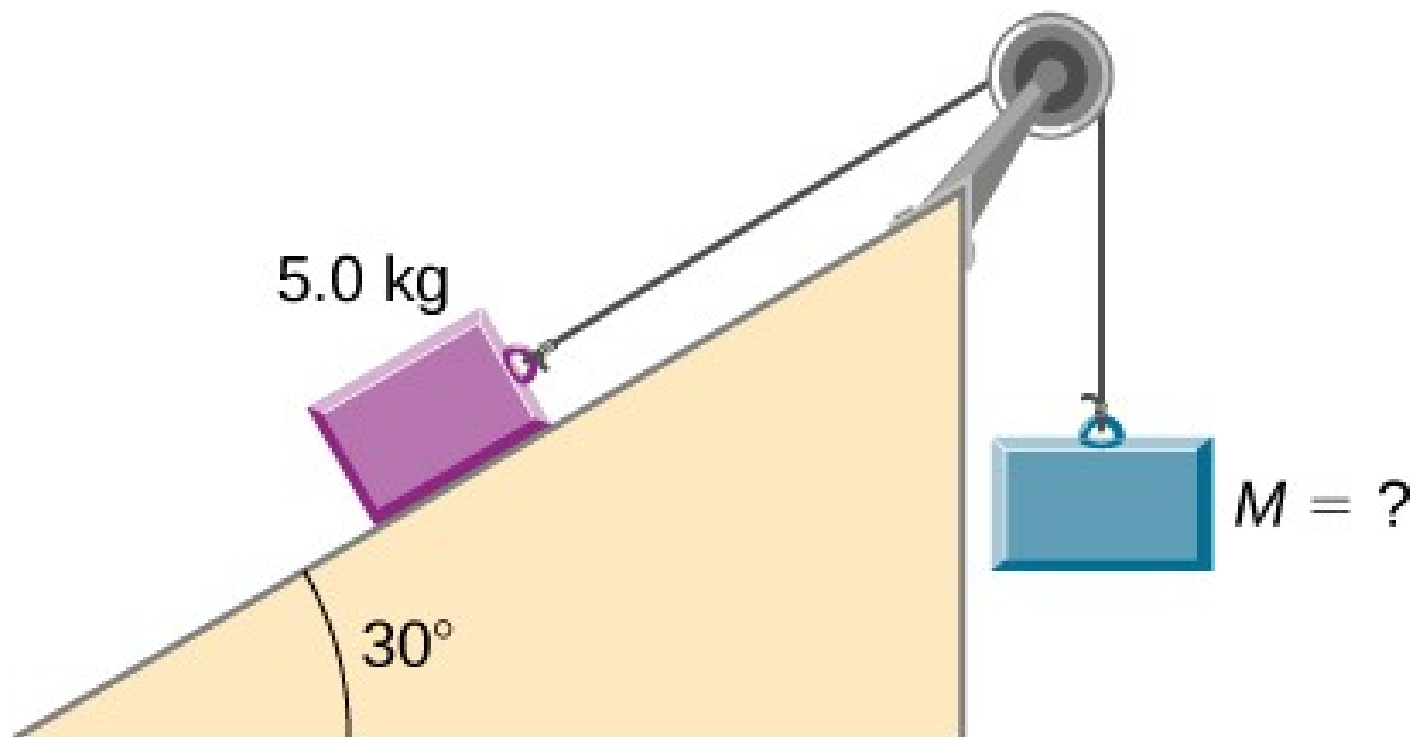
A solid sphere of radius 10 cm is allowed to rotate freely about an axis. The sphere is given a sharp blow so that its center of mass starts from the position shown in the following figure with speed 15 cm/s. What is the maximum angle that the diameter makes with the vertical?

## EXERCISE 71



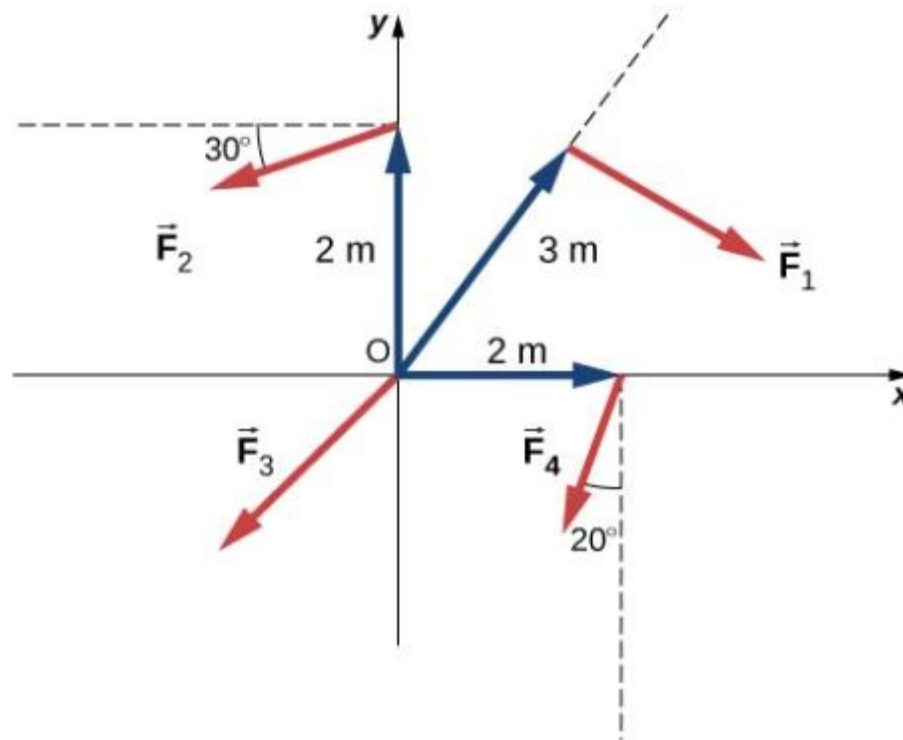
Two flywheels of negligible mass and different radii are bonded together and rotate about a common axis (see below). The smaller flywheel of radius 30 cm has a cord that has a pulling force of 50 N on it. What pulling force needs to be applied to the cord connecting the larger flywheel of radius 50 cm such that the combination does not rotate?

## EXERCISE 75



What hanging mass must be placed on the cord to keep the pulley from rotating (see the following figure)? The mass on the frictionless plane is 5.0 kg. The inner radius of the pulley is 20 cm and the outer radius is 30 cm.

## EXERCISE 77



Calculate the torque about the z-axis that is out of the page at the origin in the following figure

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